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MODULE 1

Introduction to Earth System Models

Learning outcomes

1. Define key terminology in climate modelling.
2. Explain the difference between weather prediction and climate modelling.
3. Describe the structure of a typical Earth System Model.
4. Distinguish between resolved and parameterized processes, and explain how they effect model uncertainty.

1.1 Course motivation

Climate models are among the most powerful and widely used tools in the Earth sciences. They are our most important tool for making future projections about Earth's climate (Figure 1.1), and provide a laboratory where we can safely run experiments to understand the workings of the climate system.
To critically evaluate and appropriately apply model output, users need a working understanding of:

- how models are **structured**;
- how processes are **represented**;
- what the **limitations** of the model are.

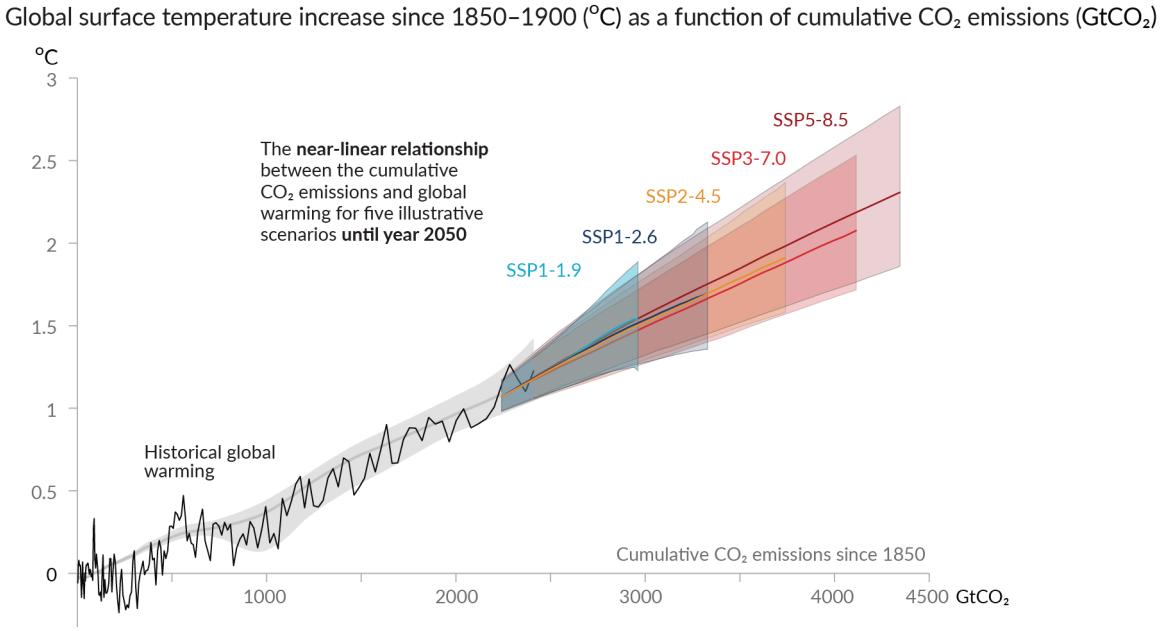


Figure 1.1: Global surface temperature increase since 1850–1900. *Figure SPM.10 in IPCC (2021).*

1.2 Weather vs climate

Distrust in weather predictions is common, so if short range weather prediction is so frequently flawed, how can we have confidence in climate projections up to a century into the future?
The question arises from a misunderstanding of the difference between weather and climate, and in turn how the two are calculated (Figure 1.2).

1.2.1 Initial value problem

Weather forecasting is best understood as an **initial value problem**. Initial value problems are common in physics, eg "if I throw a rock at speed x and angle y, where will it land?" Just like how dynamics equations are used in this example, given an initial state of the atmosphere, the governing equations of weather are integrated forward in time to produce a forecast.
The weather equations are chaotic, meaning they are extremely sensitive to their initial conditions. Two atmospheric states that differ imperceptibly at the start will diverge at roughly an exponential rate. Incredibly small errors compound rapidly, so forecasts become less accurate the further forward you go. This places a hard time limit on the accuracy of a prediction, no matter how good the model is.

1.2.2 Boundary forcing problem

Climate modelling is better characterized as a **boundary forcing problem**. Rather than moving through a particular sequence of weather, climate models are concerned with the statistical properties of the climate system, such as mean values and variability.

A particularly helpful analogy is a balloon tethered in front of a blowing fan. You cannot say where it will be in ten seconds, but you know where it is confined to because of things like the length of the string and the direction and strength of the fan. Weather is the balloon's position and climate is the area it could be. Change the fan and this shifts with it,even though the actual location continues changing.
Similarly, climate models started from varied initial conditions will produce different specific weather events, but should produce the same climatological statistics.

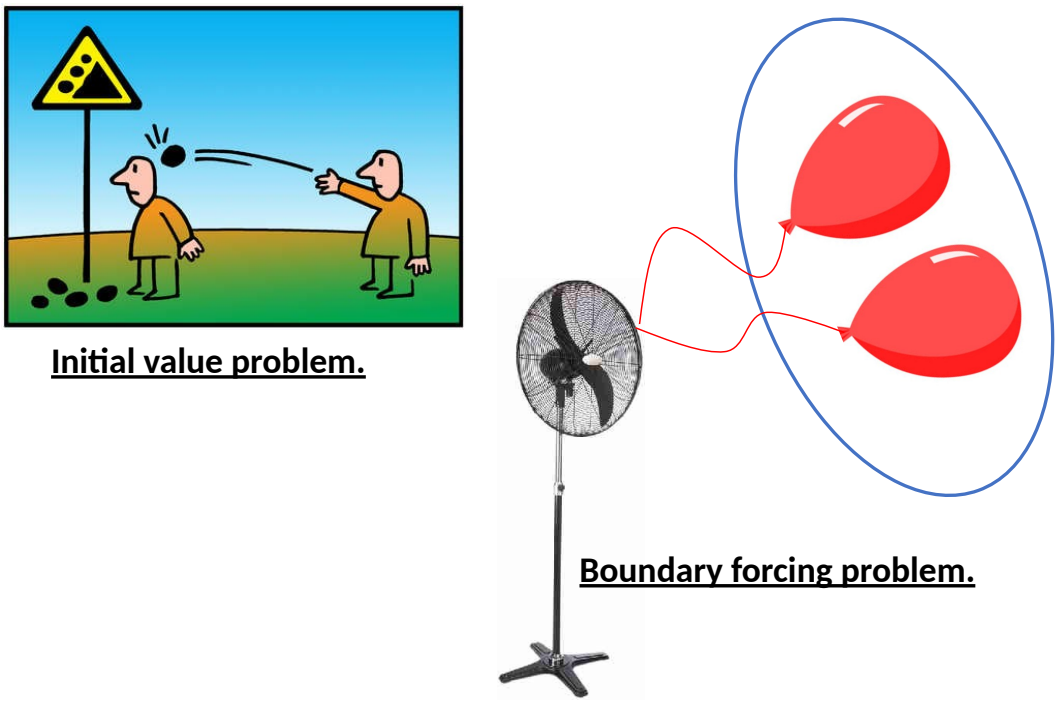


Figure 1.2: *Left:* weather forecasting as an initial value problem. *Right:* climate as a boundary forcing problem.

1.3 Human-forced and natural climate variability

Climate naturally varies across a range of different timescales. Increased volcanic activity, for example, can lower global average temperature for a period of a few years. This deviation from the trendline is relatively small, and tends to correct itself once the eruptions slow. With the addition of human interaction we see a different kind of forcing, where global temperature increases but does not return to baseline. Humans produce greenhouse gases at a steady rate, meaning the variance from the mean continues to build rather than self correcting.
Comparing a simulation run without anthropogenic forcing to one with it gives a clear picture of global warming trends, and of what is causing them (Figure 1.3).

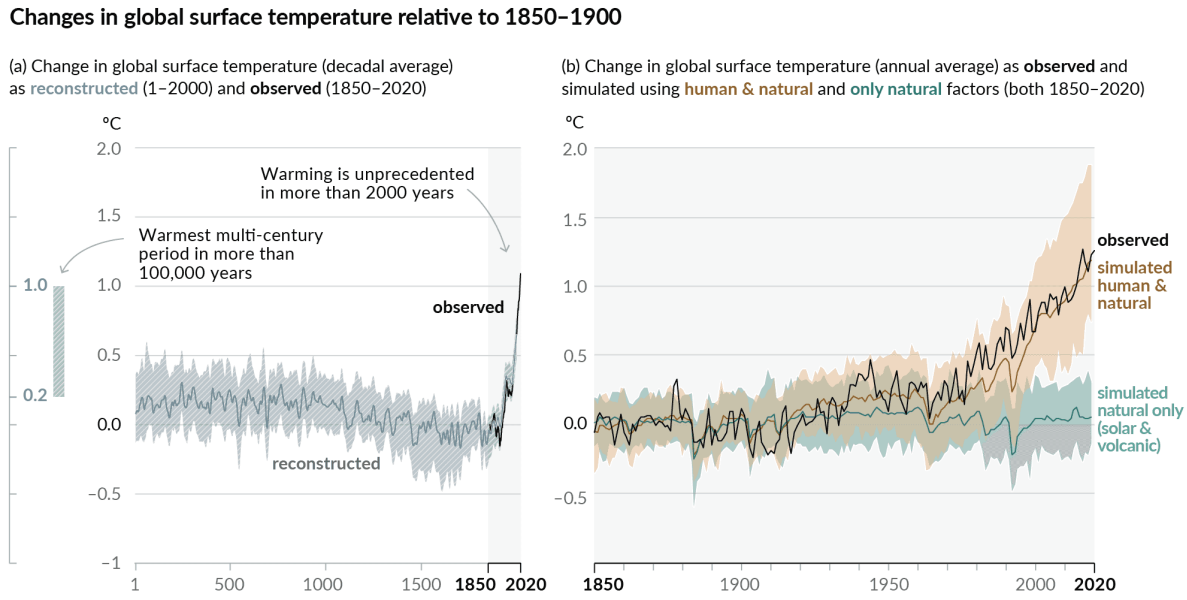


Figure 1.3: Changes in global surface temperature relative to 1850–1900. *Figure SPM.1 in IPCC (2021).*

Video: the dog is the weather

The walking-the-dog animation. The owner's steady path is the climate and the dog's erratic path around it is the weather. The dog's position at any time is unpredictable, but the direction the pair is travelling is not. *From: TeddyTV for NRK, animation by Ole Christoffer Haga.*

1.4 Key definitions and terminology

Physics-based vs data-driven modelling

A *physics-based* model derives its behaviour from the governing equations of fluid motion, thermodynamics and radiative transfer. A *statistical* or *data-driven* model takes the opposite approach. It looks at relationships between climate variables and steps them forward. **ESMs are physics-based models.**

Climate model vs GCM vs ESM

These terms are often used interchangeably, but describe different specifics.

- **Climate model** covers a wide range, from a simple energy balance model with a handful of equations to a full scale global simulation.
- A **General Circulation Model (GCM)** specifically calculates the motion of the atmosphere and/or ocean using the equations of motion.
- An **Earth System Model (ESM)** is a GCM with additional Earth system components coupled in, such as a carbon cycle, dynamic vegetation, and atmospheric chemistry. This adds more systems to feed into each other.

Dynamics vs physics

Dynamics refers to the equations governing fluid motion. *Physics* refers to everything else a model needs in order to represent radiation, cloud formation, precipitation, etc.

Resolved vs parameterized processes

A process is *resolved* if the model's grid spacing and time step are fine enough to represent it directly from the governing equations. A process is*parameterized* if it occurs at a scale smaller than the grid, and must instead be approximated using a simplified relationship.

Computing requirements

An ESM solves a large system of coupled partial differential equations at every grid point, for every time step, across several coupled models. Running one is computationally demanding and is a major research constraint.

1.5 Structure of a typical Earth System Model

1.5.1 Components and the coupler

Earth system processes can be categorized into a number of "spheres" (atmosphere, land, ocean, sea ice, land ice, etc.). These spheres have sometimes intersected throughout climate modelling history, and new ones are still developing, but the categorization is helpful for understanding the structure of an ESM.

Each component sphere is represented by a **dedicated model**. These models are fully realized and can be run independently. A researcher interested only in ocean currents can run just the ocean model. Running components on their own is useful for decreasing computational needs, and for isolating sources of model error, since any bias is a product of only that component. The component models are then linked together using a **coupler** to create the full scale ESM. The coupler passes energy flux, momentum and mass between components (Figure 1.4).

1.5.2 Towers of boxes full of warm, moist air

Climate models begin by breaking the Earth up into a massive grid (Figure 1.5)

Each grid square can be thought of as a full **column** containing all the different processes that happen at different altitudes. Almost as if you took a snapshot of the Earth and extracted a giant core starting at the center, all the way out into the atmosphere. This column is further split into boxes that encompass different layers of soil/earth and the processes they contain. The climate model then solves a system of equations to track "stuff" in the boxes:

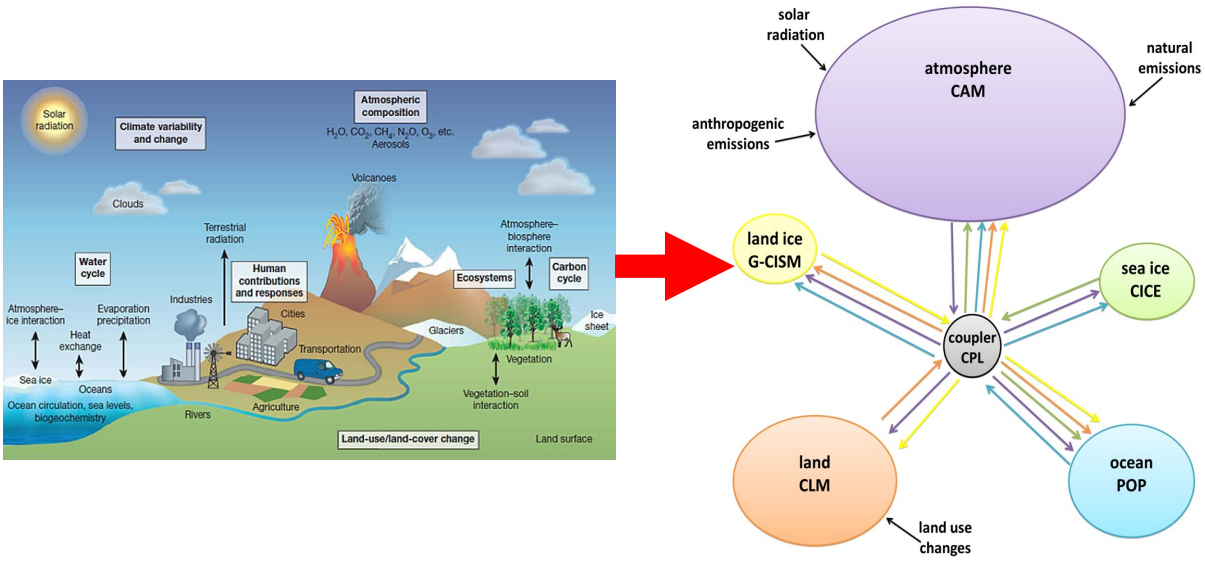


Figure 1.4: *Left:* the processes in an ESM. *Right:* a set of component models exchanging fluxes through the coupler. *Left: Figure 2.5 in Subcommittee on Global Change Research (2003). Right: adapted from Alexander and Easterbrook (2015).*

- energy
- momentum
- water (3 phases)
- greenhouse gases
- ozone
- aerosols and particulates
- biogeochemistry
- vegetation (land)

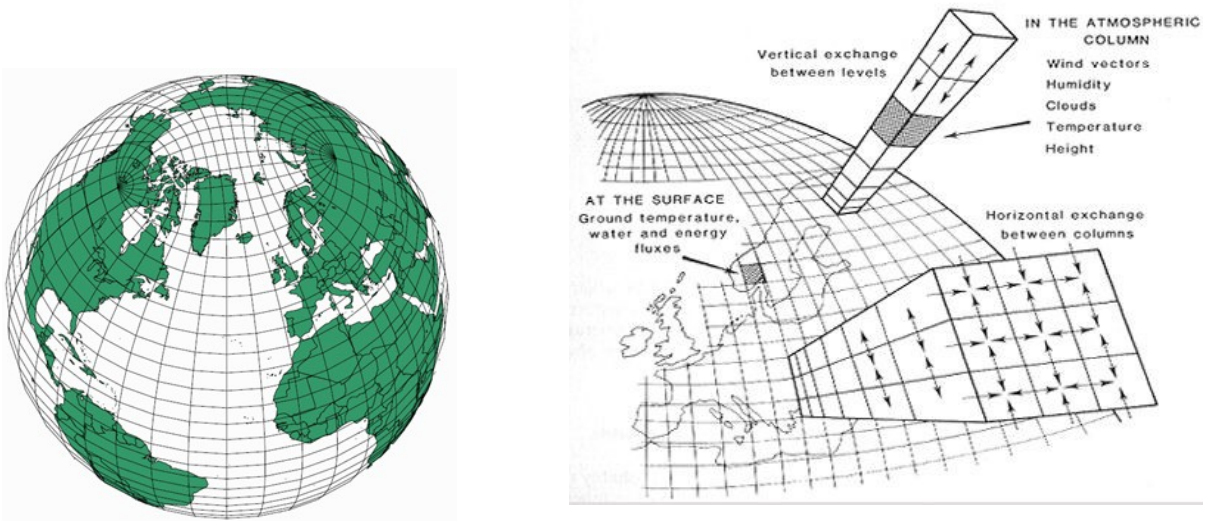


Figure 1.5: The model grid. *Left:* the horizontal mesh laid over the globe. *Right:* each surface cell is the base of a vertical column of boxes.

An ESM is *not* a statistical model based on past data. Nothing in its formulation requires it to have observed the climate.

1.6 Resolved and parameterized processes

1.6.1 Resolved processes

The processes that climate models handle most reliably are the ones that operate at a large scale than one grid cell, and that are governed by known laws of physics.

Large-scale atmospheric and oceanic circulation falls into this category. Incoming solar radiation and outgoing longwave radiation is another example, since this exchange follows radiative transfer physics (Figure 1.6).

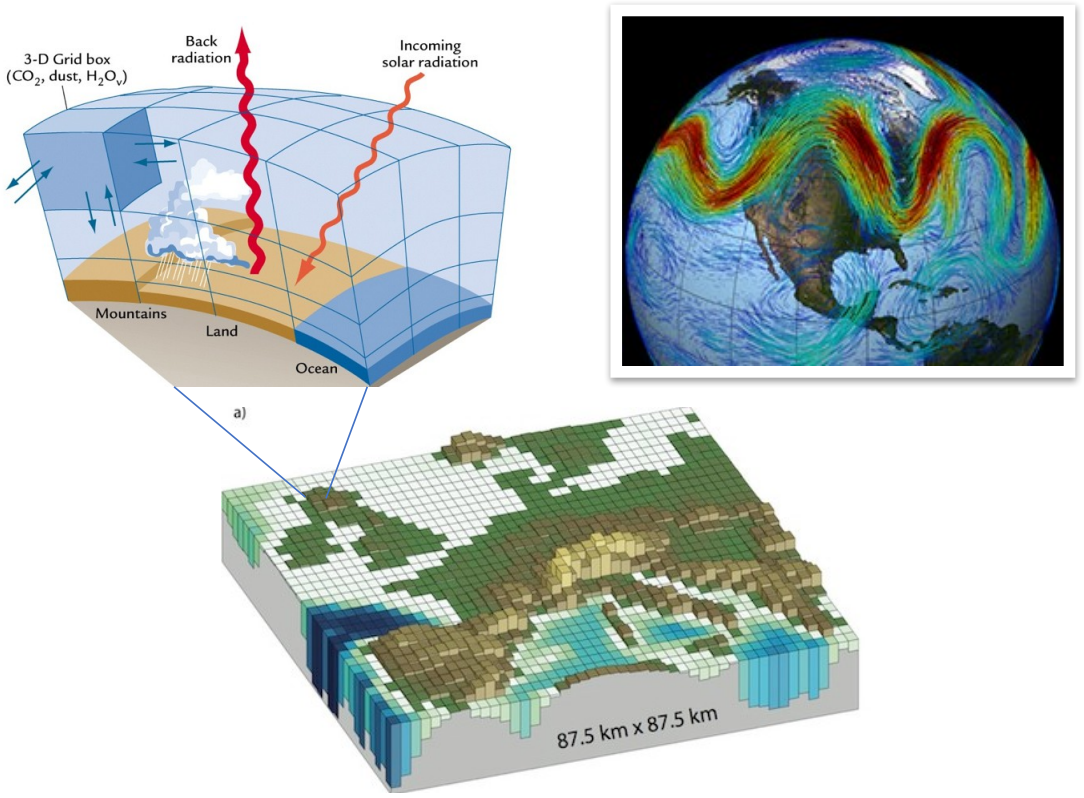


Figure 1.6: Resolved processes.

1.6.2 The governing equations

The resolved dynamics of the atmosphere are described by a set of coupled non-linear PDEs expressing conservation of momentum, mass, energy and water (Figure 1.7).

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p - 2\boldsymbol{\Omega} \times \boldsymbol{u} + \boldsymbol{g} + \boldsymbol{F}, \tag{1.1}$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{1.2}$$

$$c_p \frac{DT}{Dt} - \frac{1}{\rho}\frac{Dp}{Dt} = Q, \tag{1.3}$$

$$\frac{Dq}{Dt} = S_q, \tag{1.4}$$

$$p = \rho R T. \tag{1.5}$$

where

$\boldsymbol{u}$ velocity vector

ρ density

p pressure
$\boldsymbol{\Omega}$ Earth's rotation vector, giving the Coriolis term
$\boldsymbol{g}$ effective gravity
$\boldsymbol{F}$ friction and unresolved momentum transport
T temperature
c_p specific heat capacity at constant pressure
Q diabatic heating, including radiation and latent heating
q specific humidity
S_q moisture sources and sinks
R gas constant for dry air

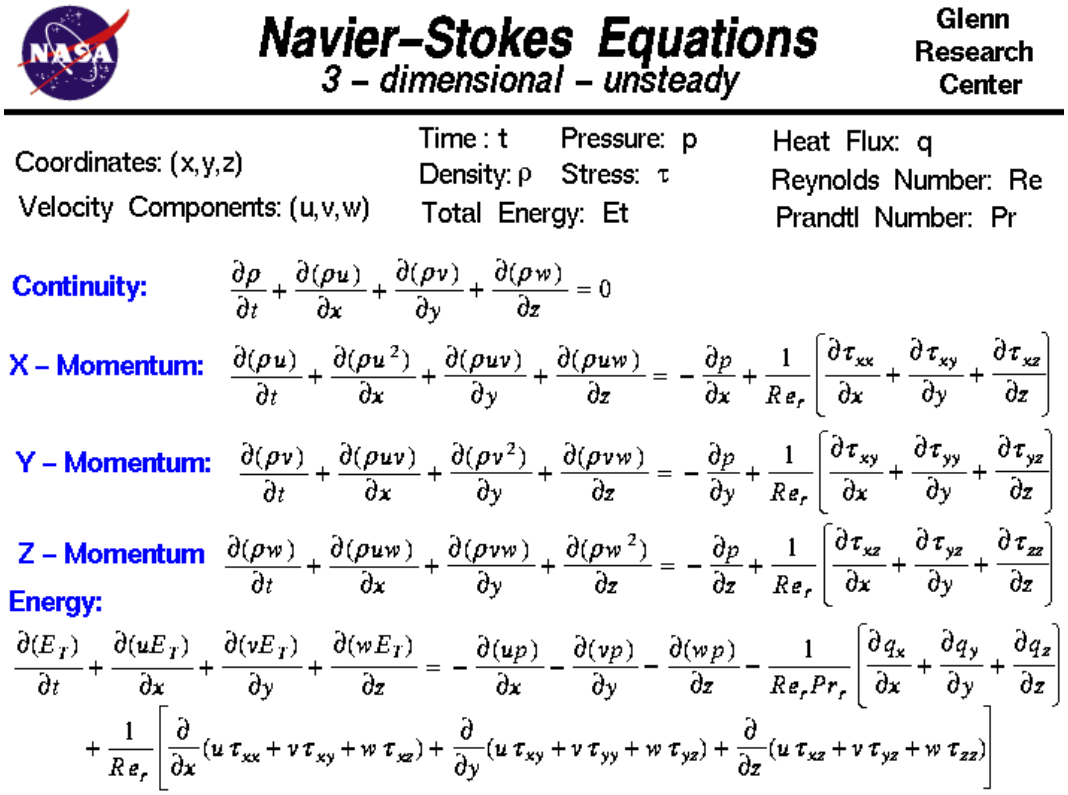


Figure 1.7: The same conservation statements as they are usually written in fluid dynamics, in flux form. *From: NASA Glenn Research Center, Beginner's Guide to Aeronautics.*

The conservation statements themselves are exact. The terms $\boldsymbol{F}$, Q and S_q are where parameterization schemes come in. This is the main source of differences between models and the highest concentration of modern research takes place here.

1.6.3 Unresolved processes: parameterization

Processes that are too complex, or too small-scale, to be represented by the governing equations in a grid box are **parameterized** (Figure 1.8). This involves developing an empirical relation between the small-scale process and the grid box scale.Grid boxes can be relatively large, sometimes with sides of100 km. A great deal therefore falls below the resolved scale: convection, cloud microphysics, turbulence, gravity wave drag. These parameterizations are **physically based, but limited by available observations**.



Figure 1.8: Processes that fall below the grid scale and must therefore be parameterized. *From: The COMET Program*.

1.7 The evolution of Earth System Models

Over the years, what we now know as ESMs have become more and more sophisticated. More spheres have been added as research has continued. The term "Earth System Model" itself came into use to reflect the increased complexity by around 2010, and this complexity is still increasing (Figure 1.9). Note that this added complexity has not necessarily narrowed the uncertainty which we will touch on more later.

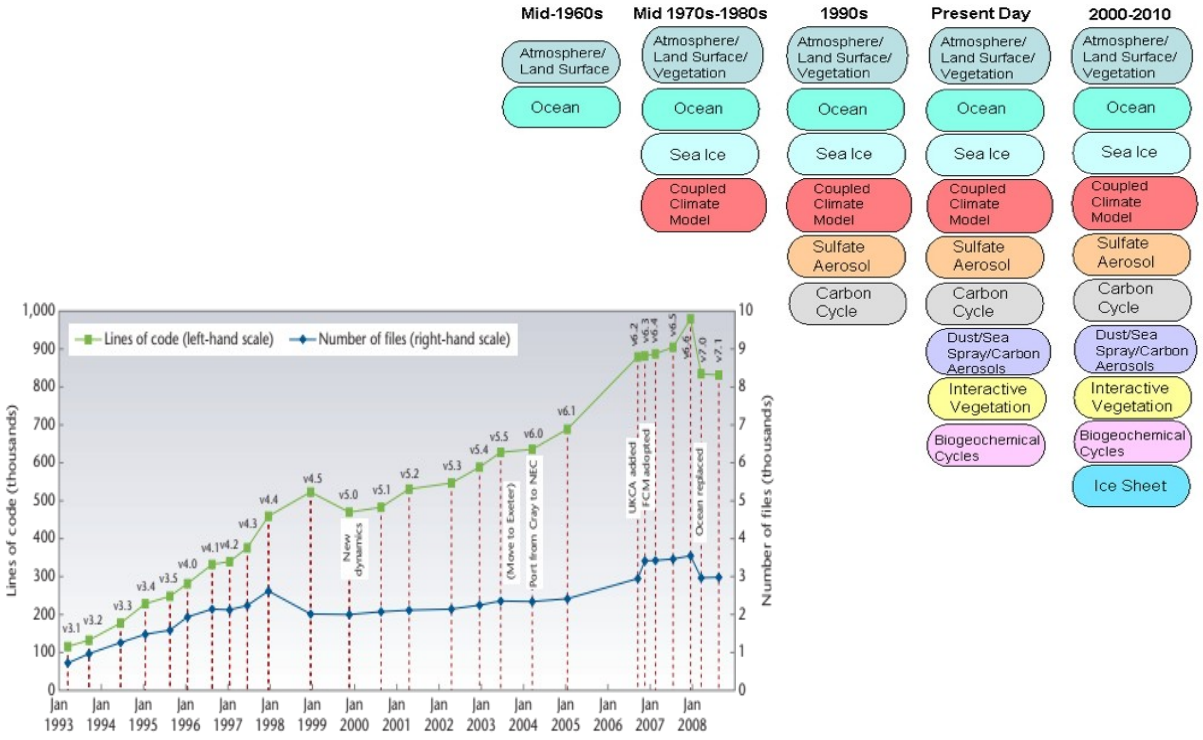


Figure 1.9: Growth of an Earth system model over time. *From: Easterbrook and Johns (2009).*

Further reading

For a full history of ESMs and how they came to be, see this presentation by Steve Easterbrook: https://prezi.com/pakaaiek3nol/timeline-of-climate-modeling/

MODULE 2

Radiative Forcing and Climate Sensitivity

Learning outcomes

1. Distinguish between the different definitions of radiative forcing.
2. Derive the no-feedback temperature response from a zero dimensional energy balance model.
3. Distinguish between the different measures of climate sensitivity.
4. Explain the difficulties involved in estimating climate sensitivity from the historical record.
5. Evaluate the argument that climate sensitivity may not be a useful metric for guiding policy.

2.1 The global energy budget

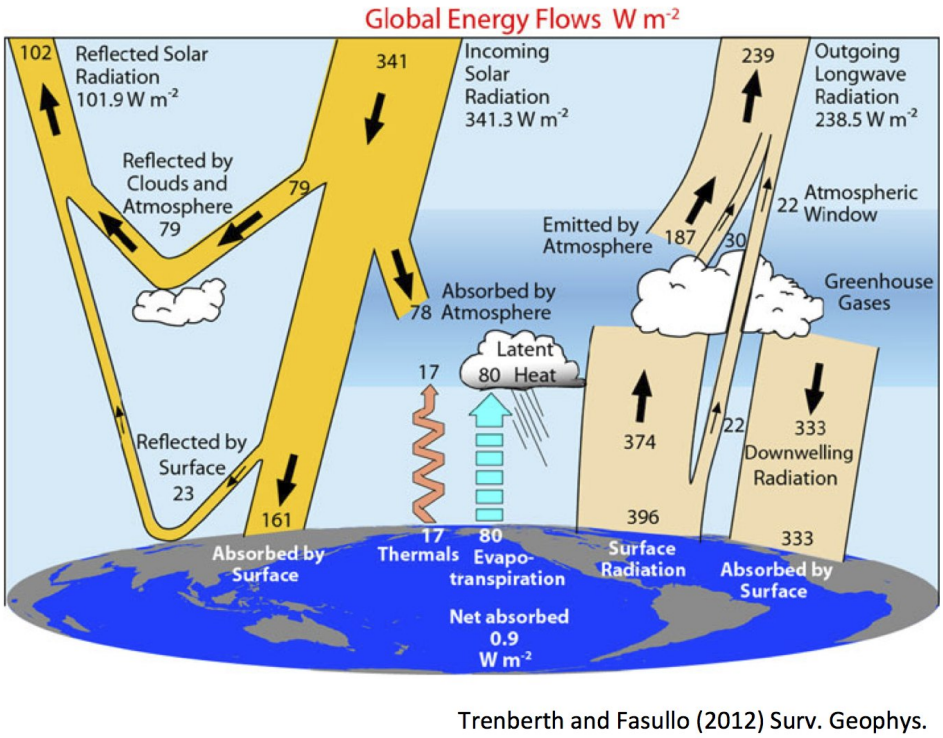


Figure 2.1: Global energy flows in $\mathrm{W\,m^{-2}}$. *From: Trenberth and Fasullo (2012).*

The climate system sits in an energy balance (Figure 2.1). There is shortwave radiation incoming from the sun and longwave radiation being emitted by the earth. Not all of the incoming solar is absorbed. Roughly 30 % is reflected by clouds, the atmosphere and the surface, and that fraction is the **planetary albedo**. The surface emits considerably more longwave radiation than actually escapes to space. Greenhouse gases and clouds absorb most of it and re-emit in both directions. The downward part is what keeps the surface far warmer than the absorbed solar alone could sustain.

Radiation is not the only way energy leaves the surface. Thermals and evapotranspiration carry a substantial share upward, so the surface budget cannot be closed with radiation alone. The budget does not balance exactly. The small residual is the energy currently accumulating in the climate system, most of which goes into the ocean.

2.2 Radiative forcing

Radiative forcing (RF) is an imposed change to this balance. Adding CO_2 reduces the outgoing longwave for a given surface temperature, which leaves a positive imbalance at the top of atmosphere. The system removes that imbalance by warming until the outgoing longwave rises to match the absorbed solar again.

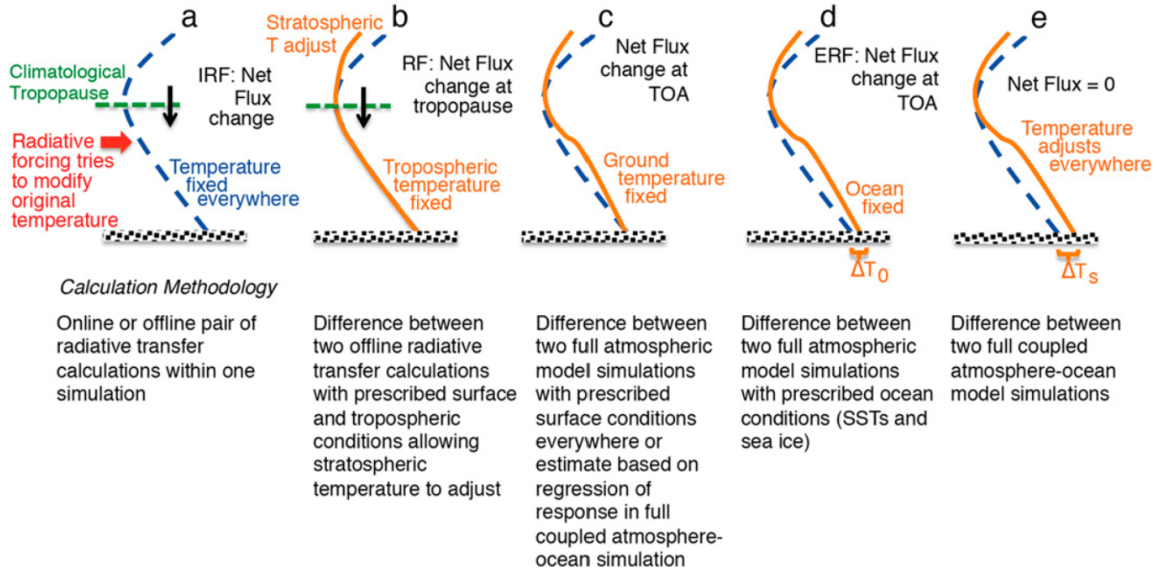


Figure 2.2: The definitions of radiative forcing compared, with the calculation method each one uses below it. *Figure 8.1 in IPCC (2014).*

The RF concept assumes that change in net irradiance in response to the forcing can be separated from the subsequent responses which occur after the forcing is applied. These subsequent responses are called feedbacks and will be gone over in more detail in the next module. In reality, decomposing RF and feedbacks requires a number of assumptions. Different definitions of RF make these assumptions in various ways that include or don't include different responses (Figure 2.2).

Instantaneous RF (IRF)

The instant change in the net radiative flux at the top of the tropopause (TOA) in response to forcing change. The instantaneous RF is **not an accurate predictor of the temperature response**, as many relatively rapid, temperature invariant changes take

place in the stratosphere and troposphere (known as adjustments).

Stratospherically Adjusted RF (SARF)

Accounts for stratospheric adjustments in response to forcing agents. For example, the stratosphere will undergo cooling in response to increasing CO_2. The SARF does **not** account for tropospheric adjustments, or other temperature invariant adjustments.

Takes into account both stratospheric and tropospheric adjustments that occur in response to the forcing. The ERF is the definition most frequently used in the AR6.

It is the change in the net TOA radiative flux in response to a forcing agent, after accounting for adjustments in both stratospheric and tropospheric temperatures, water vapour, clouds and aerosols and some aspects of Earth's albedo which are temperature independent (such as vegetation induced changes),

The key here is that adjustments occur before changes in Earth's surface temperature, and is separate from the surface temperature response. Feedbacks, on the other hand, are defined as changes that scale with surface temperature, which is how they enter Equation (2.7).

2.3 Climate sensitivity and feedbacks

Climate sensitivity refers to the global mean surface temperature increase associated with a doubling of pre-industrial CO_2, i.e. an atmospheric concentration of 560 ppm.

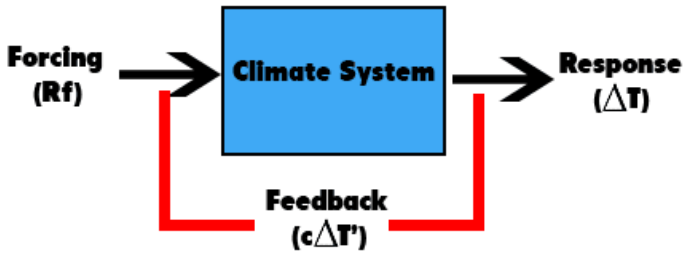


Figure 2.3: The climate system as a forcing–response system with feedback.

Estimating a value for CS is particularly challenging, however, because a large part of the total climate response is driven by feedbacks (Figure 2.3).**Feedbacks** are physical changes to the climate system that arise in response to the original warming caused by increasing CO_2. The magnitude of some feedbacks is uncertain due to a combination of lack of available observations and limitations in the model representation of key processes, such as clouds.

2.3.1 Theoretical climate response in absence of feedbacks

From a 0D energy balance model, we can derive the climate sensitivity parameter from an equilibrium temperature state like we talked about earlier. Differentiating the Stefan-Boltzmann law gives the **Planck feedback parameter**, which is the rate at which outgoing longwave radiation increases per unit warming:

$$\lambda_{\mathrm{P}} = 4\sigma T_{eq}^3 \approx 3.3\,\mathrm{W\,m^{-2}\,K^{-1}}. \tag{2.1}$$

The **climate sensitivity parameter** λ_0 is connected to radiative imbalance and is defined by:

$$\lambda_0 = \frac{1}{4\sigma T_{eq}^3}. \tag{2.2}$$

where

σ Stefan–Boltzmann constant
T_{eq} equilibrium (effective emission) temperature
λ_P Planck feedback parameter, in $\mathrm{W\,m^{-2}\,K^{-1}}$
λ_0 climate sensitivity parameter, in $\mathrm{K\,m^2\,W^{-1}}$

Our best estimate of $\lambda_0 = \sim 0.3\,\mathrm{K\,m^2\,W^{-1}}$. Interested readers can see the derivation of this equation, and an explanation, in Brian Rose's ClimLab:

- lecture on the 0D EBM, https://brian-rose.github.io/ClimateLaboratoryBook/courseware/zero-dim-ebm/
- analytical solution of the 0D EBM, https://brian-rose.github.io/ClimateLaboratoryBook/courseware/analytical-efolding/

Temperature response is a function of the climate sensitivity parameter and the radiative forcing:

$$\Delta T_0 = \lambda_0 \, \Delta R_f, \tag{2.3}$$

$$\Delta T_0 = 0.3\,\mathrm{K\,m^2\,W^{-1}} \times 3.9\,\mathrm{W\,m^{-2}} \approx 1.2\,\mathrm{K}. \tag{2.4}$$

ΔR_f is the radiative forcing, or net energy imbalance at the top of the atmosphere, for a doubling of CO_2. Best estimate of $\Delta R_f = 3.9\,\mathrm{W\,m^{-2}}$.

2.3.2 Climate response with feedbacks

$$\Delta T = \lambda_0 (\Delta R_f + c\,\Delta T'). \tag{2.5}$$

Actual response of the climate system will also include feedbacks. Because the magnitude of feedbacks tends to be less agreed upon, this heavily effects the accepted range of equilibrium climate sensitivity and makes it hard to constrain (Table 2.1).

Source	**ECS range**
Charney (1979) Report (Charney et al., 1979)	1.5 – 4.5 K
CMIP3	2.1 – 4.4 K
CMIP5	2.1 – 4.7 K
CMIP6	1.8 – 5.6 K

Table 2.1: Model based estimates of equilibrium climate sensitivity (ECS). CMIP5 and CMIP6 ranges from Zelinka et al. (2020).

2.4 Different flavours of climate sensitivity

Similar to RF, there are multiple ways to define climate sensitivity, including the transient climate response (TCR), the equilibrium climate sensitivity (ECS), and the effective climate sensitivity (EffCS) (Figure 2.4). The TCR, ECS and EffCS explore the warming response on different timescales, and consequently include differences in their assumptions about which climate feedbacks are included.

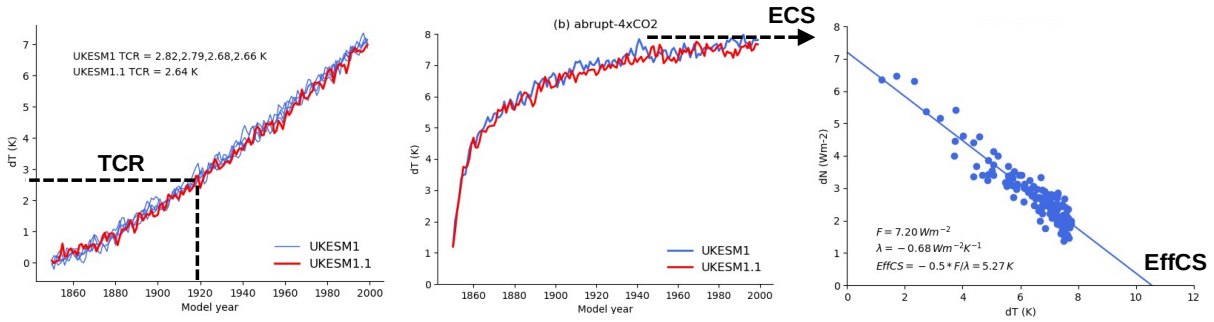


Figure 2.4: Three examples of model-based measures of climate sensitivity. *Left:* the TCR from a 1% yr^{-1} scenario. *Centre:* the ECS from a $4\times CO_2$ run. *Right:* EffCS from a Gregory regression of the TOA net flux N against warming. *From: Mulcahy et al. (2023).*

Equilibrium Climate Sensitivity (ECS)

Most commonly used definition of climate sensitivity. ECS is defined as the long-term global mean warming expected from a doubling of atmospheric CO_2 (relative to preindustrial concentrations), after allowing the ocean to reach an equilibrium state.

Transient Climate Response (TCR)

The global mean temperature change (relative to the pre-industrial period) at the time when CO_2 doubles, usually under a $1\,\%\,\mathrm{yr}^{-1}$ experiment.
It will **always be less than the ECS**, because it doesn't account for many of the longer term climate feedbacks.

Cess climate sensitivity

Calculated from climate model experiments where a sea surface temperatures (SSTs) are increased by 2 K in a pre-industrial control simulation, i.e. the climate system is left unperturbed from preindustrial conditions beyond the prescribed SST perturbation.

$$\lambda_{\mathrm{Cess}} = \frac{\Delta N}{\Delta T_s}. \tag{2.6}$$

The *Cess Feedback*, λ_{Cess}, is then calculated as the net change in the global mean TOA radiation, ΔN, that evolves from global mean surface temperature change, ΔT_s, which arises from the perturbed SSTs.
The global TOA radiation budget can be approximated as the sum of the radiative forcing, F, and a surface temperature feedback term $\lambda\Delta T$:

$$N = F + \lambda\,\Delta T. \tag{2.7}$$

At equilibrium the radiation is in balance and $N = 0$. Hence we can estimate the Cess Sensitivity as a function of the radiative forcing from a doubling of atmospheric CO_2, $F_{2\times}$, and the Cess feedback term:

$$\Delta T_{\mathrm{Cess}} = \frac{-F_{2\times}}{\lambda_{\mathrm{Cess}}}. \tag{2.8}$$

Effective Climate Sensitivity (EffCS)

An estimate of the climate sensitivity obtained from a Gregory et al. (2004) type regression of the change in temperature against the net change in the TOA radiative flux.
It is a common way to estimate the ECS from a model simulation prior to the system reaching equilibrium, assuming the magnitude of net feedbacks remains constant. Often **smaller than the true ECS**, since it underestimates the effective radiative forcing.

$$\mathrm{ECS} = \frac{-F_{2\times \mathrm{CO}_2}}{\lambda}. \tag{2.9}$$

where

$F_{2\times \mathrm{CO}_2}$ adjusted radiative forcing at TOA (the y-intercept) for a doubling of CO_2
λ feedback parameter (the slope of the regression)

Earth System Sensitivity (ESS)

Includes very long term Earth system feedbacks, including the response of ice sheets and long term vegetation changes, on millennial time scales, which occur long after the ocean has reached an equilibrium state.
Note that the ESS is beyond the capability of most or all modelling centres to estimate using ESMs.

2.5 Estimating climate sensitivity from observational evidence

Two main routes have been used, and both get compared against model estimates: the historical temperature record, and paleoclimate records.

2.5.1 Historical Record

When using the Historical record, estimates of radiative forcing and feedbacks come from historical runs of a climate model (see Dessler, 2020). The climate system is not in equilibrium over the historical period, so adding up the individual feedbacks can leave a large residual. A modified approach is usually needed, splitting the radiative forcing into clear sky and cloudy sky parts. Three difficulties follow:

1. Internal variability matters. The warming pattern we observed is only one of many that could have happened, known as the**unforced pattern effect**.
2. The observed warming pattern is not the same as the pattern at equilibrium with $2 \times \mathrm{CO}_2$, because the system has not caught up with the forcing yet. This is the **forced pattern effect**.
3. Aerosol forcing over the historical period is poorly constrained, and feedbacks depend on the climate state, so their strength over the historical period need not match their strength at equilibrium.

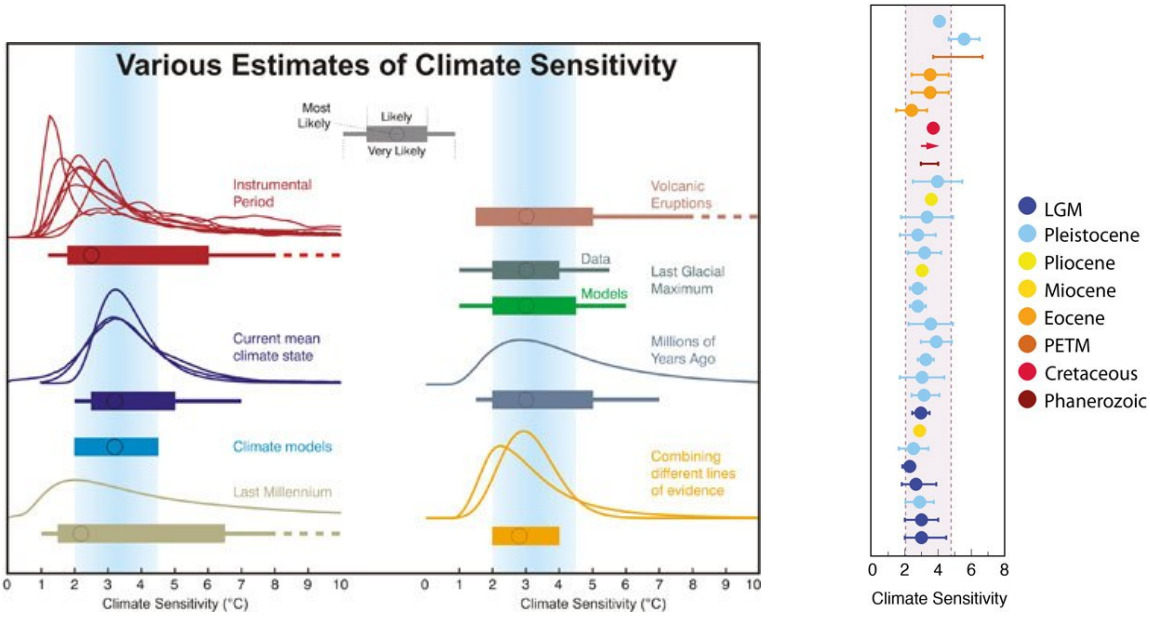
2.5.2 Paleoclimate Evidence

Useing the paleoclimate evidence uses time periods thought to be reasonable analogues for present day change. This requires estimating the difference $\Delta\lambda$ between the feedback parameter at $2 \times \mathrm{CO}_2$ and the one in the paleoclimate state. That difference then enters the ECS estimate:

$$\mathrm{ECS} = \frac{-F_{2\times \mathrm{CO}_2}}{\lambda_{\mathrm{paleoclimate}} + \Delta\lambda}. \tag{2.10}$$

It is worth using more than one paleo record, since the pattern of warming can differ a lot between periods even at similar greenhouse gas concentrations, because vegetation, ice sheet extent and continental configuration were all different.

2.6 Estimating CS: putting it all together



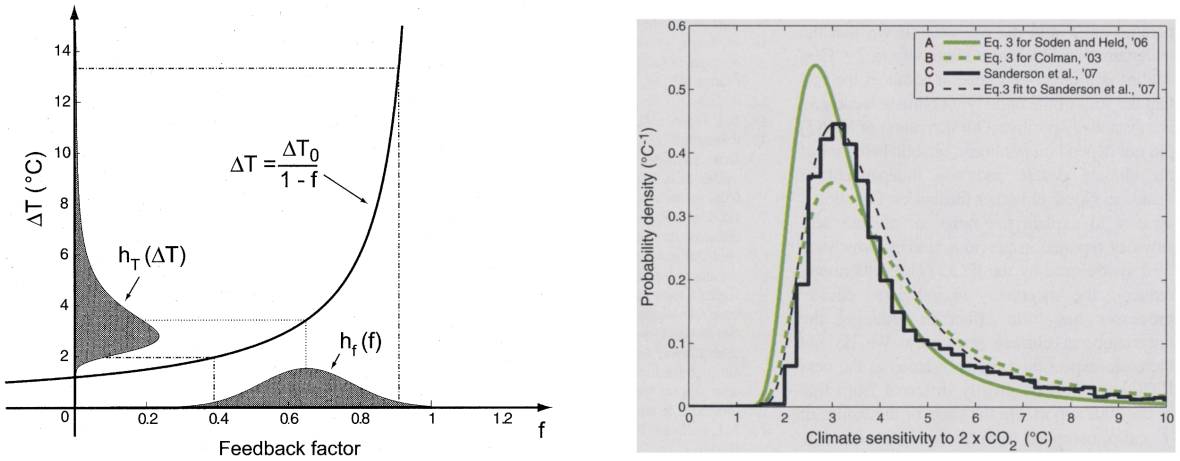
(a) (b)

Figure 2.5: Climate sensitivity estimates. *(a)* Estimates grouped by source *(b)* Paleoclimate estimates alone, coloured by period. *Adapted from Knutti and Hegerl (2008) and PALAEOSENS Project Members (2012) by John Cook, Skeptical Science.*

2.7 Is climate sensitivity even a useful metric?

Despite nearly 50 years of research since the Charney Report (Charney et al., 1979), it has been difficult to constrain the estimates of climate sensitivity (Figure 2.5).

Adding further complexity to models, for example, has not improved our understanding of the true ECS, and has in many ways increased the range from the original estimates given in the Charney report. Many of the Earth System components in today's climate models (such as the carbon cycle, and dynamic ice sheets) are the most uncertain. We've also seen an increase in the number of modelling centres participating in model intercomparison projects, each with differences in how model physics are represented, and the degree of complexity included (Figure 2.6).



(a) (b)

Figure 2.6: Why the upper bound on climate sensitivity is inherently hard to find. *(a)* The response $\Delta T = \Delta T_0/(1-f)$ against the feedback factor f. *(b)* The sensitivity distributions that follow from several published feedback estimates. *From: Roe and Baker (2007).*

Allen and Frame, Science, 2007

"An upper bound on the climate sensitivity has become the holy grail of climate research. As Roe and Baker point out, it is inherently hard to find. It promises lasting fame and happiness to the finder, but it may not exist and turns out not to be very useful if you do find it. Time to call off the quest." (**?**)

We know from the research that has been done to date that emission reductions are urgently needed. While the actual emissions reductions required to reach a particular temperature target will depend on the true climate sensitivity, we can already make informed policy decisions using the information we have available to us now. Waiting until we know what the true climate sensitivity will be before making policy decisions on climate change is too late.

MODULE 3

Climate Feedbacks

Learning outcomes

1. Define a climate feedback in radiative terms.
2. Break the net feedback parameter down into its classical components.
3. Explain why the timescale on which a feedback operates determines whether an ESM can assess it.
4. Describe the snow albedo feedback.
5. Explain why cloud feedbacks remains the main source of spread in CS estimates.

3.1 Theoretical Basis

Climate feedbacks

The surface temperature response incorporates a wide range of climate feedbacks which amplify (positive feedbacks) or reduce (negative feedbacks) the climate response to a radiative forcing.
Climate feedbacks represent the change in radiative flux per unit change in temperature for a given process ($\mathrm{W\,m^{-2}\,K^{-1}}$). They are **emergent properties** of the climate system, and are therefore difficult to validate.

3.1.1 Feedback timescales

Feedbacks operate over a variety of different timescales, from years to millennia (Figure 3.1). Typically we can only assess feedbacks operating over years to decades in climate models, which are considered **fast feedbacks**. These are the feedbacks typically included in estimates of the ECS we obtain from ESMs. These are thought to provide an additional $1.0-3.5\,\mathrm{K}$ of warming above the $1.2\,\mathrm{K}$ we assessed from a 0D energy balance perspective (the no feedback case). Other feedbacks, which operate over centuries to millennia (**slow feedbacks**), are usually not testable in ESMs because they require very long simulations, though some of these have been explored in simpler models such as Earth System Models of Intermediate Complexity (EMICs). From more simplified EMICs, our best guess of the Earth System Sensitivity (ESS), after these slow feedbacks are accounted for, lies somewhere between $3\,\mathrm{K}$ and $6\,\mathrm{K}$.

3.1.2 Net Feedback Equation

Recall from Equation (2.4) that $2\times CO_2$ without feedbacks gives $\Delta T\sim 1.2\,\mathrm{K}$. The impacts of individual climate feedbacks, λ_x, can be estimated as a function of the change in the net TOA radiative flux per unit change in x ($\partial N/\partial x$), and the change in variable x as a function of the surface temperature change (dx/dT) (Soden and Held, 2006):

$$\lambda_x = \frac{\partial N}{\partial x}\,\frac{dx}{dT}. \tag{3.1}$$

The net feedback parameter, λ, is typically assumed to be a linear sum of the individual radiative feedbacks which we will identify in the follow section:

$$\lambda = \lambda_{\text{planck}} + \lambda_{\text{LR}} + \lambda_{\text{RH}} + \lambda_{\text{a}} + \lambda_{\text{C,SW}} + \lambda_{\text{C,LW}}. \tag{3.2}$$

Key idea

Note: Other feedback mechanisms exist as well (such as permafrost carbon, vegetation feedbacks, marine carbon cycle feedbacks, and others), though they are not typically assessed when using the classical feedback decomposition so we will not be covering them here.

3.2 Feedback Term Breakdown

3.2.1 Planck feedback, λ_{planck}

Starting with λ_{planck}, this term is from the Stefan-Boltzmann response of the surface and atmosphere. It follows the convention that a warmer planet emits more longwave radiation to space. It is by far the largest single term, and it is the stabilizing feedback that keeps the climate system bounded. It tends to be well constrained across models (high levels of agreement), since it follows directly from radiative transfer and largely does not contribute to spread in output across models.

3.2.2 Lapse rate feedback, λ_{LR} and Water vapour feedback, λ_{RH}

In the tropics, moist convection warms the upper troposphere faster than the surface, which increases emission to space relative to a uniform warming. Lapse rate feedback, λ_{LR}, captures the change in radiative flux due to this nonuniform change. The sign is often negative, but is positive at high latitudes, where a stable boundary layer concentrates warming near the surface, contributing to Arctic amplification. It should be noted that when adding the term over all regions, it does come out to be a net negative.

Water vapour feedback, λ_{RH}, quantifies the result of water vapour in the air. Largely based on the Clausius-Clapeyron relationship, warmer air holds more water vapour, and water vapour is itself a strong greenhouse gas leading to λ_{RH} being the largest positive term. Models agree closely on it when relative humidity is assumed to stay roughly constant.

Lapse rate feedback and water vapour feedback are anti-correlated terms. When both are combined into one and analyzed across multiple models it tends to be well constrained, but a larger spread is found when analyzing each alone (Soden and Held, 2006).

Key idea

The remaining terms, λ_{a}, $\lambda_{\text{C,SW}}$, and $\lambda_{\text{C,LW}}$ account for a large portion of disagreement between models and thus will be examined in a bit more depth.

3.2.3 Surface albedo feedback, λ_{a}

Snow Albedo Feedback (SAF) defined as the change in net shortwave radiation as a function of the change in surface air temperature:

$$\frac{\partial Q}{\partial T_s} = -I \frac{\partial \alpha_p}{\partial \alpha_s} \frac{\Delta \alpha_s}{\Delta T_s}. \tag{3.3}$$

where

Q net SW radiation at the surface
I incoming SW at TOA
α_p planetary albedo
α_s snow albedo

T_s surface air temperature

It is driven by snow cover loss and sea ice decline. These processes expose darker land and ocean surfaces, which reduces the reflection of sunlight. SAF is concentrated at high latitudes and in the seasonal snow zones.

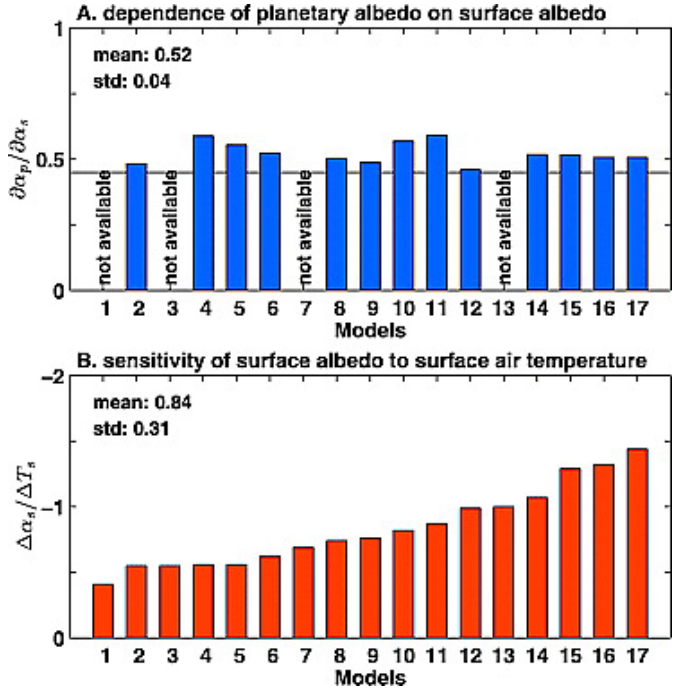


Figure 3.2: The two factors of the snow albedo feedback, evaluated across 17 CMIP3 models. *(A)* The radiative transfer term $\partial\alpha_p/\partial\alpha_s$ *(B)* The surface term $\Delta\alpha_s/\Delta T_s$. *Figure 1 in Hall and Qu (2006).*

Where the SAF uncertainty actually sits

As seen in figure 3.2, SAF is often split into two factors: *(A)* The radiative transfer term $\partial\alpha_p/\partial\alpha_s$ *(B)* The surface term $\Delta\alpha_s/\Delta T_s$. Splitting the feedback like thisd is useful because it localizes where the disagreement between models actually sits.

- The radiative transfer term is well constrained.
- Essentially all of the inter-model spread is in the surface term, which is also the term we can observe over the seasonal cycle.

Qu and Hall (2007) noted that the total SAF (NET) could be separated into a component based on the reduction in albedo in response to changes in snow cover (SNC), and a term related to the change in the albedo of the snowpack due to thermal metamorphosis or temperature (TEM), assuming that:

1. S and α_{snow} are solely functions of air temperature, and

2. α_{land} is independent of temperature.

$$\mathrm{NET} = \mathrm{SNC} + \mathrm{TEM}, \tag{3.4}$$

$$\mathrm{SNC} = (\alpha_{snow} - \alpha_{land}) \frac{\Delta S}{\Delta T}, \tag{3.5}$$

$$\mathrm{TEM} = S \frac{\Delta \alpha_{snow}}{\Delta T}. \tag{3.6}$$

where

α_{snow} snow covered albedo
α_{land} snow-free albedo
S snow-covered fraction
T surface air temperature

Qu and Hall (2007) and Fletcher et al. (2015) found that in CMIP3 models:

- SNC generally accounts for a greater proportion of the total SAF than TEM over most regions, while **TEM was more important over the Arctic**.
- SNC generally **overestimated** over midlatitudes, while **underestimated** by most models over the Arctic.
- TEM **underestimated** by most models over midlatitudes, and **overestimated** over the Arctic.

Fletcher et al. (2015) found similar relationships in CMIP5, which suggests these are structural biases rather than artefacts of a single model generation (Figure 3.3).

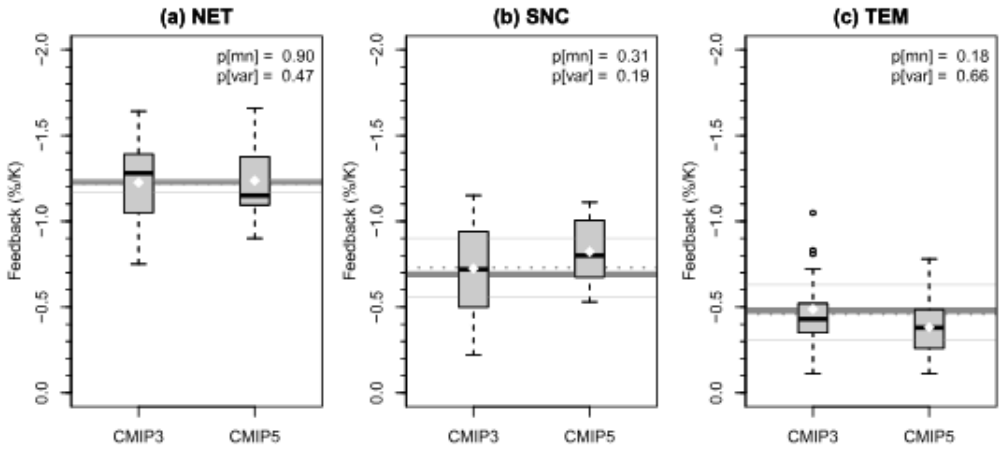


Figure 3.3: Spread of the MAMJ-mean SAF terms across CMIP3 and CMIP5 models. *From: Fletcher et al. (2015).*

3.2.4 Cloud feedbacks, $\lambda_{\mathrm{C,SW}}$ and $\lambda_{\mathrm{C,LW}}$

Cloud feedback is defined as the sum of the feedbacks from individual cloud regimes, each weighted by the fractional global coverage of that regime (the "bottom-up" approach of IPCC AR6, Forster et al., 2021):

$$\lambda_{\mathrm{C}} = \sum_{r} \lambda_{\mathrm{C,r}} = \lambda_{\mathrm{high\,alt}} + \lambda_{\mathrm{high\,amt}} + \lambda_{\mathrm{low}} + \lambda_{\mathrm{land}} + \lambda_{\mathrm{mid}} + \lambda_{\mathrm{OD}} + \lambda_{\mathrm{Arctic}}. \tag{3.7}$$

where

$\lambda_{\text{high alt}}$ high-cloud altitude feedback
$\lambda_{\text{high amt}}$ tropical high-cloud amount feedback
λ_{low} subtropical marine low-cloud feedback
λ_{land} land cloud feedback
λ_{mid} mid-latitude cloud amount feedback
λ_{OD} extratropical cloud optical depth feedback
λ_{Arctic} Arctic cloud feedback

Clouds reflect sunlight (SW, cooling) and absorb surface emission to re-emit it at a lower temperature (LW, warming). Their net effect is the cloud radiative effect (CRE), the difference in TOA radiation between all-sky and clear-sky conditions. Each regime responds differently to warming, so the net cloud feedback is the sum of competing SW and LW effects.

Cloud feedbacks in CMIP6

The net cloud feedback is positive, but cloud feedbacks are also a main reason CMIP6 models have a wider and higher ECS range than CMIP5 (Zelinka et al., 2020). We will cover this a bit more in Section 4.5, but it is good to note early that regimes labeled as "low confidence" effect this ECS spread the most.

AR6 found the following for each regime (Forster et al., 2021):

- **High-cloud altitude (positive).** Cloud tops rise with the tropopause, and under the *fixed anvil temperature* mechanism their temperature stays roughly constant. LW emission therefore does not increase as the surface warms, strengthening the high-cloud greenhouse effect. Supported by theory, satellite records and cloud-resolving models.
- **Tropical high-cloud amount (negative, low confidence).** Under the *stability iris* mechanism, anvil cloud area decreases as upper-tropospheric stability increases, and aggregation dries the surrounding air, increasing LW emission. Part of this is offset by reduced SW reflection. Models are biased and depend on convective parametrization.
- **Subtropical marine low cloud (positive).** Reduced low-cloud amount reflects less sunlight, with stratocumulus dominating over trade cumulus. Uncertainty has fallen considerably since AR5.
- **Land cloud (positive, low confidence).** Subtropical land drying reduces low cloud, but models have biases over land and observational support is limited.
- **Mid-latitude cloud amount (positive, medium confidence).** Low-cloud fraction decreases with rising SST between 30° and 60°. The proposed link to poleward jet shifts **does not hold**, since high- and low-cloud responses cancel in net CRE.
- **Extratropical optical depth (small negative, medium confidence).** Ice-dominated clouds are replaced by optically thicker liquid clouds over the Southern Ocean. Many CMIP5 models **overestimated** this because they had too little supercooled liquid.
- **Arctic cloud (small positive, low confidence).** More low cloud in autumn and winter as sea ice declines increases downwelling LW at the surface, but the TOA effect is very small and a negative sign is almost as probable.

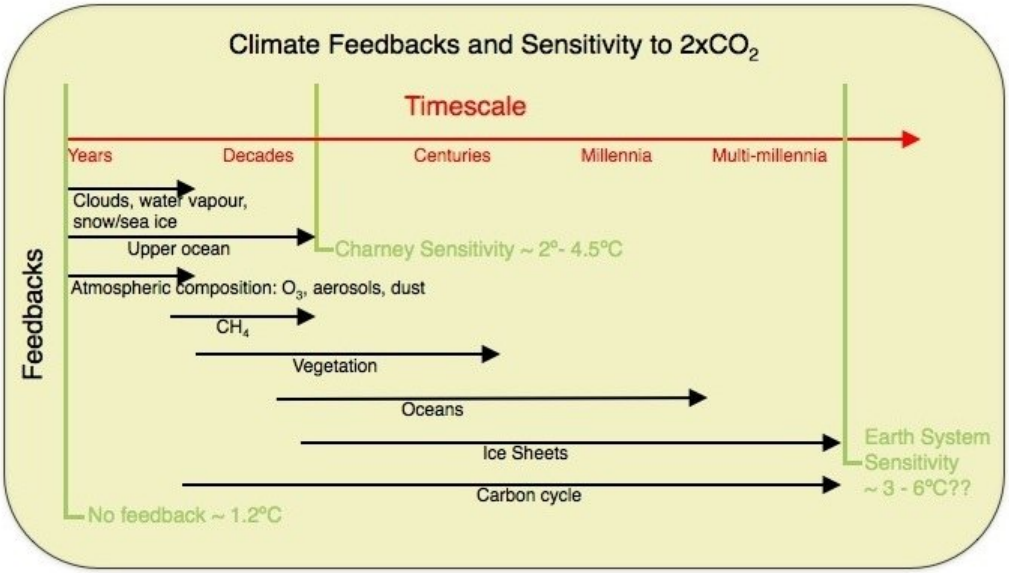


Figure 3.1: Climate feedbacks arranged by the timescale. *From: RealClimate, On sensitivity: Part I.*

MODULE 4

Multi-Model Ensembles (CMIP)

Learning outcomes

1. Understand the meaning of a model ensemble.
2. Describe the differences between the types of model ensemble, and give a general overview of their uses.
3. Understand the scientific rationale behind model ensembles and the Coupled Model Intercomparison Project (CMIP).
4. Explain how multi-model ensembles are used to quantify structural uncertainty and emission scenario uncertainty.
5. Appreciate the relative degrees of uncertainty in CMIP6 global mean temperature and precipitation projections.

4.1 What is a model ensemble?

Model Ensemble

"A collection of comparable datasets that reflect variations within the bounds of one or more sources of uncertainty and that, when averaged, can provide a more robust estimate of underlying behaviour. (IPCC, 2021) (Figure 4.1)"

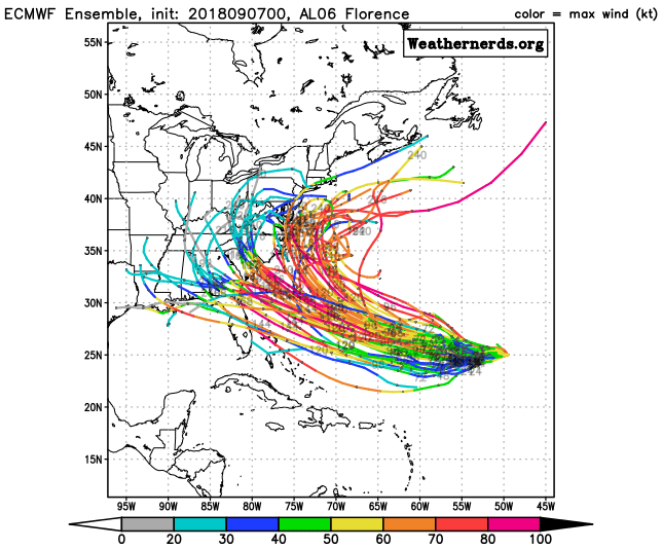


Figure 4.1: *ECMWF* ensemble forecast tracks for Hurricane Florence, 00Z 7 September 2018. *From: Penn State METEO 3 - Lesson 13.*

4.1.1 The three types of model ensembles

The overall uncertainty of a forecast or projection arises from *multiple sources of uncertainty*, and different types of model ensembles are used to assess different kinds of uncertainty. Here, we distinguish between three main types of model ensembles: single model intial condition large ensembles (SMILEs), multi-model ensembles, and perturbed parameter ensembles (Table 4.1, Figure 4.2).

Type	Construction	Samples
Single model initial-condition large ensemble (SMILE)	One model, many runs, small perturbations to the initial state	Internal variability
Multi-model ensemble (e.g. CMIP)	Many different models, standardised experiments	Structural and scenario uncertainty
Perturbed parameter ensemble (PPE)	One model, parameters varied across plausible ranges and combinations	Parametric uncertainty

Table 4.1: The three ensemble designs, and the source of uncertainty each is built to isolate. SMILEs are the subject of Module S1 and PPEs of Module S2.

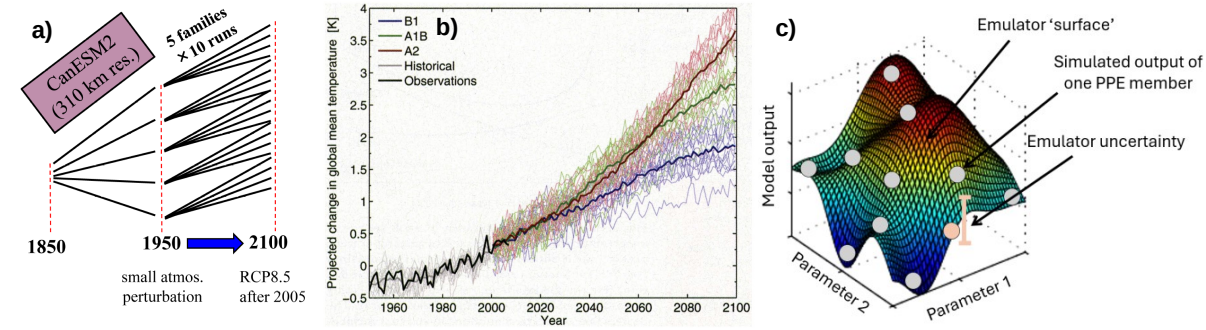


Figure 4.2: The three types of model ensemble.

4.1.2 Matching the ensemble to the uncertainty

Different kinds of model ensembles are used to assess different types of uncertainty. In this module you will be introduced to the different types of uncertainty associated with climate projections, while Module 6 will discuss ways to quantify their relative contributions.

Internal variability

Natural variability arising in the absence of external forcing, from processes intrinsic to the atmosphere, ocean, land, and cryosphere and their interactions (explored in Module S1).
Assessed using: initial-condition ensembles from a single model (SMILEs).

Structural uncertainty

Uncertainty arising from assumptions in model physics and from model resolution. (See Figure 4.3)
Assessed using: multi-model ensembles. This often requires comparing model output against historical observations to quantify a model's *bias* — the magnitude and direction

of its error. Note the catch: there is **no guarantee that the model with the lowest historical bias will produce the most realistic future projection**. Different combinations of model errors can yield the same historical skill, a problem known as **equifinality**.

Parametric uncertainty

Uncertainty arising from poorly constrained model parameters, particularly in cloud and aerosol schemes (explored in Module S2).
Assessed using: perturbed parameter ensembles.

Scenario (emissions) uncertainty

Uncertainty arising from our inability to predict future economic decisions, global and regional socioeconomic patterns, land use change, and mitigation choices (See Figure 4.4)
Assessed using: multi-model ensembles run under multiple, equally plausible future scenarios — from aggressive mitigation through to continued heavy reliance on fossil fuels. Differences between scenarios grow as projections extend further into the future, so this term **increases with lead time** while the others do not.

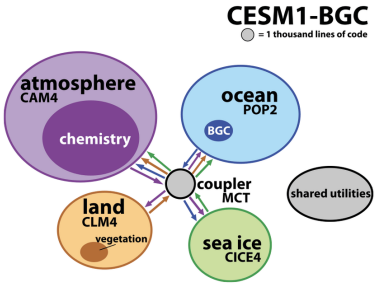


Figure 1. Architecture diagram for CESM1-BGC.

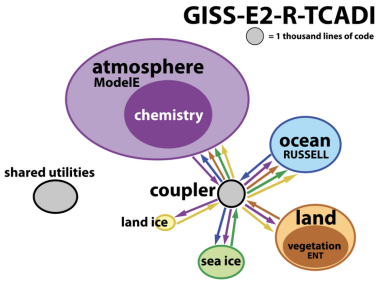


Figure 3. Architecture diagram for GISS-E2-R-TCADI.

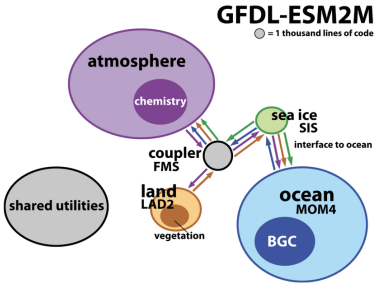


Figure 2. Architecture diagram for GFDL-ESM2M.

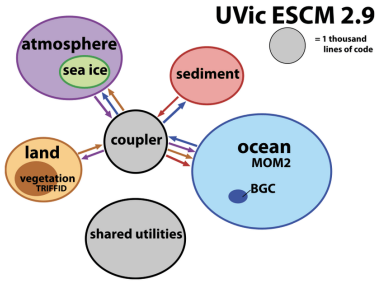


Figure 4. Architecture diagram for UVic ESCM 2.9.

(a) star-shaped

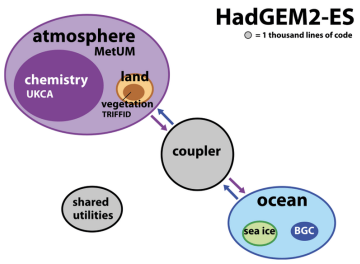


Figure 5. Architecture diagram for HadGEM2-ES.

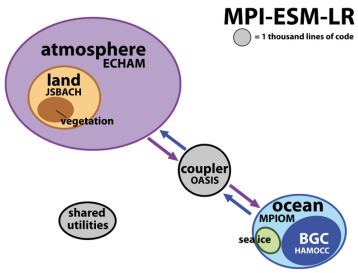


Figure 7. Architecture diagram for MPI-ESM-LR.

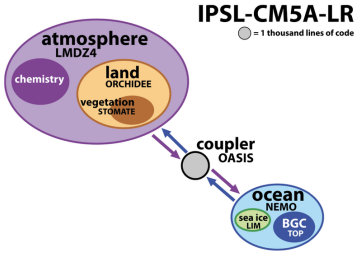


Figure 6. Architecture diagram for IPSL-CM5A-LR.

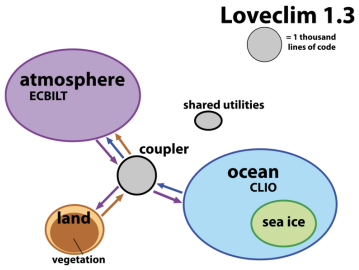


Figure 8. Architecture diagram for Loveclim 1.3.

(b) two-sided

Figure 4.3: Star-shaped and two-sided model architectures. *Adapted from Alexander and Easterbrook (2015).*

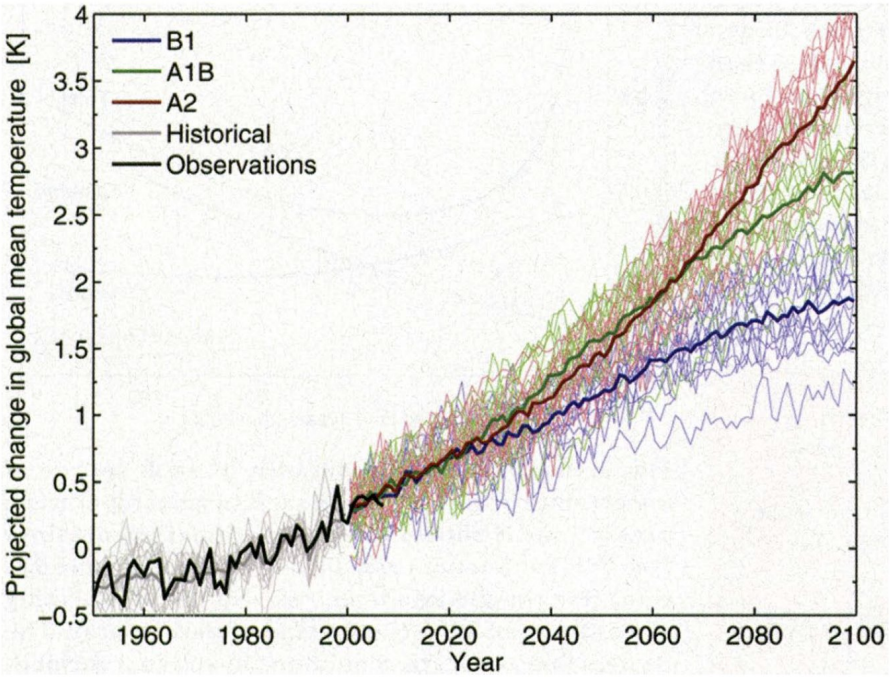


Figure 4.4: A multi-model ensemble. *Figure 1 in Hawkins and Sutton (2009).*

4.2 Scientific rationale behind model ensembles

4.2.1 History of climate model intercomparisons

The first climate model intercomparison was the Charney Report (Charney et al., 1979) (Figure 4.5), which included five models from two institutions. It aimed to address five key questions:

1. How will global warming change clouds and their radiative effects (cloud forcing and feedbacks)? Could it lead to more warming?
2. Are small aerosol particles increasing or decreasing global warming (the aerosol effect)?
3. How much CO_2 and heat from the surface will the deep ocean take up?
4. How will climate change affect different parts of Earth (spatial variability in warming)?
5. What will the future pathway of human greenhouse gas and aerosol emissions look like?

When it is assumed that the CO_2 content of the atmosphere is doubled and statistical thermal equilibrium is achieved, the more realistic of the modeling efforts predict a global surface warming of between 2°C and 3.5°C, with greater increases at high latitudes. This range reflects both uncertainties in physical understanding and inaccuracies arising from the need to reduce the mathematical problem to one that can be handled by even the fastest available electronic computers. It is significant, however, that none of the model calculations predicts negligible warming.

Figure 4.5: ECS estimates in the Charney Report (Charney et al., 1979)

The Coupled Model Intercomparison Project (CMIP) officially began in 1995, as a way to compare the ability of different GCMs to simulate historical and future climates (See Figures 4.5 - 4.8). Each CMIP uses a set of **standardised experiments** and compares model output across them, with the aim of quantifying model uncertainty. The CMIPs have historically served

as input to the IPCC Assessment Reports (Table 4.2).

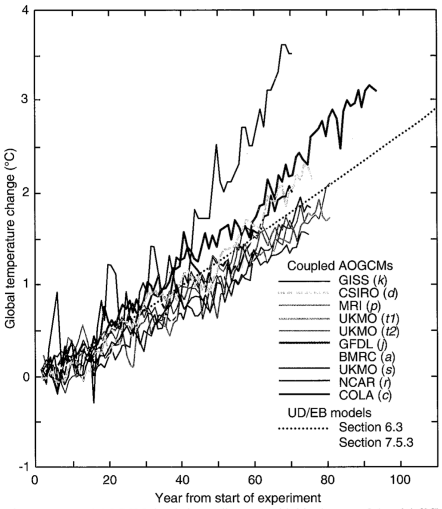


Figure 6.4: Comparison between several AOGCM simulations (climate sensitivities between 2.1 and 4.6°C), the UD/EB model of Section 6.3 (climate sensitivity 2.5°C) and the simple climate model of Section 7.5.3 (climate sensitivity of about 2.2°C). All models were forced with 1%/yr (compound) increase of atmospheric CO_2 concentration from equilibrium or near-equilibrium in 1990.

Figure 4.6: Global mean temperature projections in the IPCC SAR. *Figure 6.4 in IPCC (1996).*

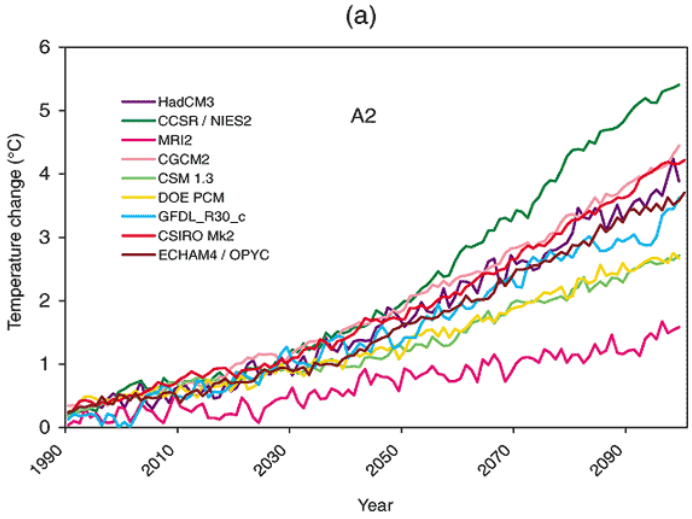


Figure 4.7: Global mean temperature projections in the IPCC TAR. *Figure 9.6a in IPCC (2001).*

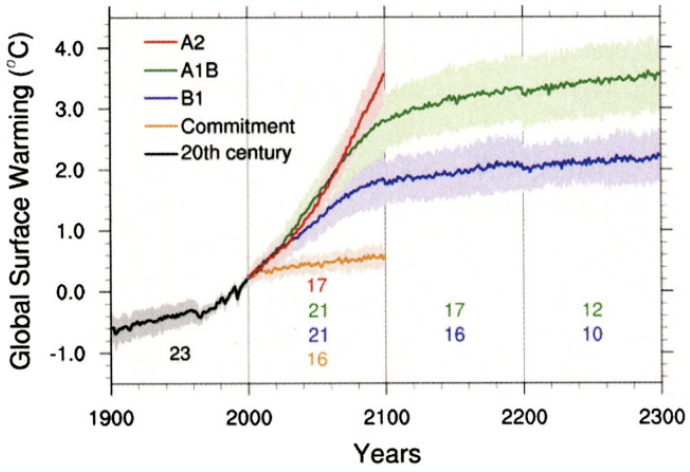


Figure 4.8: Global mean temperature projections in CMIP3. *Figure 10.4 in Meehl et al. (2007).*

CMIP phase	Period	Feeds into
—	—	IPCC FAR (1990)
CMIP1 + CMIP2	1995; 1997–2000	IPCC AR2 (1995), AR3 (2001)
CMIP3	2005–2007	IPCC AR4 (2007)
CMIP5	2008–2013	IPCC AR5 (2013)
CMIP6	2016–2021	IPCC AR6 (2023)

Table 4.2: Six CMIP phases and six IPCC Assessment Reports. Read more on the history of the CMIPs in: Touzé-Peiffer et al. (2020)

4.2.2 Ensembles of opportunity

Multi-model ensembles assume that each climate model is an *independent sample*, and that by sampling the probability space spanned by the ensemble we learn something about the true uncertainty of the system (Figure 4.9). That assumption is worth examining.

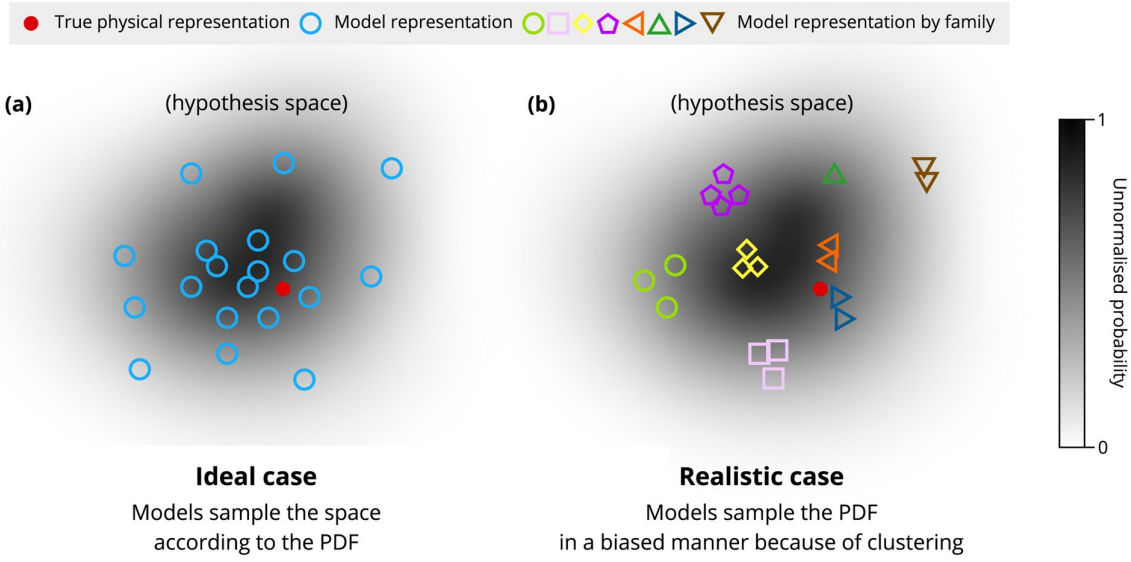


Figure 4.9: Model sample space. *Figure 1 in Kuma et al. (2023).*

Why CMIP is an "ensemble of opportunity"

Model ensembles are **ensembles of opportunity**: we use them because they happen to be what is available, not because they were designed to span the uncertainty space. Three consequences follow:

- Each individual model represents a single realization, which **underestimates** the true uncertainty within it.
- The ensemble consists only of models included in the CMIP framework.
- Models **share components** — land, ocean, or atmosphere modules are reused across centres — which clusters both the models and their results (Figure 4.10).

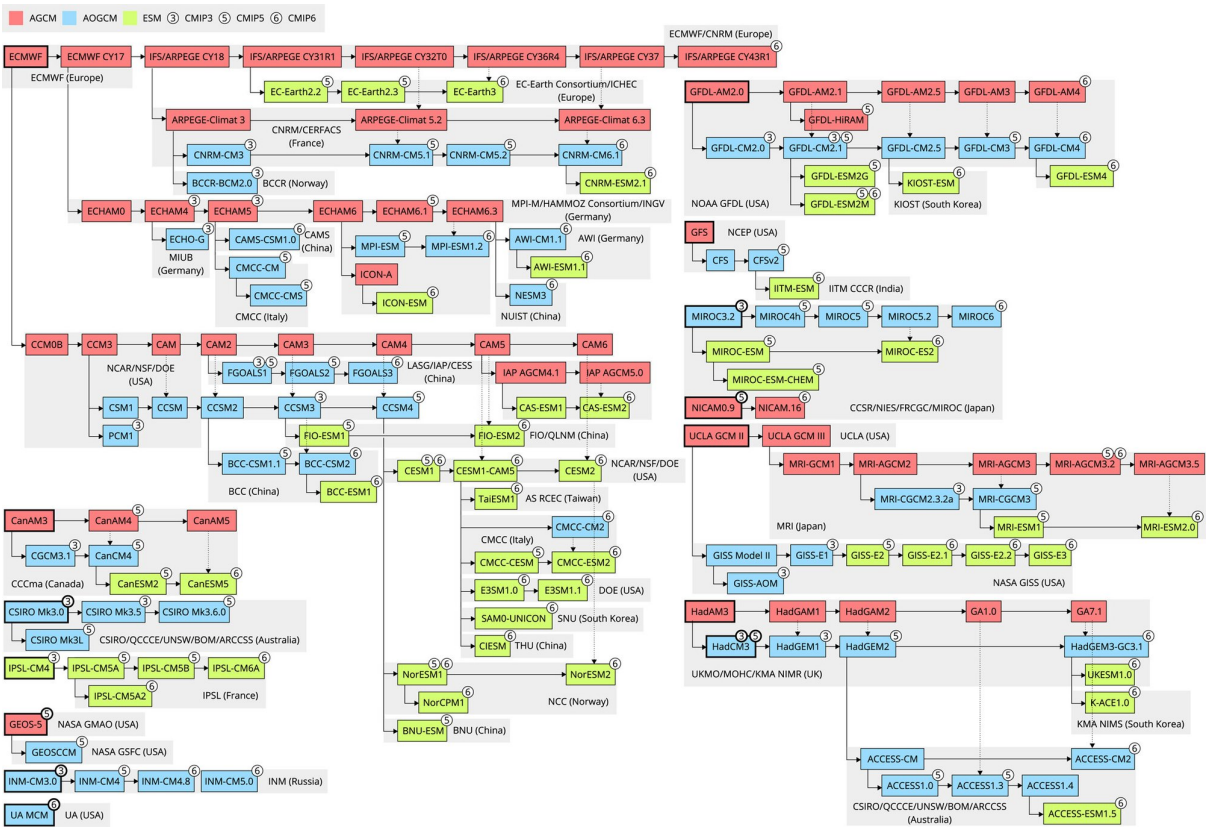


Figure 4.10: Climate model genealogy. *Figure 2 in Kuma et al. (2023).*

4.3 CMIP6

4.3.1 Overview

CMIP6 incorporated **23 separate MIPs** exploring different aspects of the climate system and different aspects of uncertainty (Figure 4.11). All participating models took part in the **DECK** (Diagnostic, Evaluation and Characterization of Klima), consisting of the following experiments:

- pre-industrial control (`piControl`) simulation;
- historical atmospheric model intercomparison (`AMIP`) simulation;
- abrupt CO_2 quadrupling simulation;
- $1\,\%\,\mathrm{yr}^{-1}$ CO_2 increase simulation.

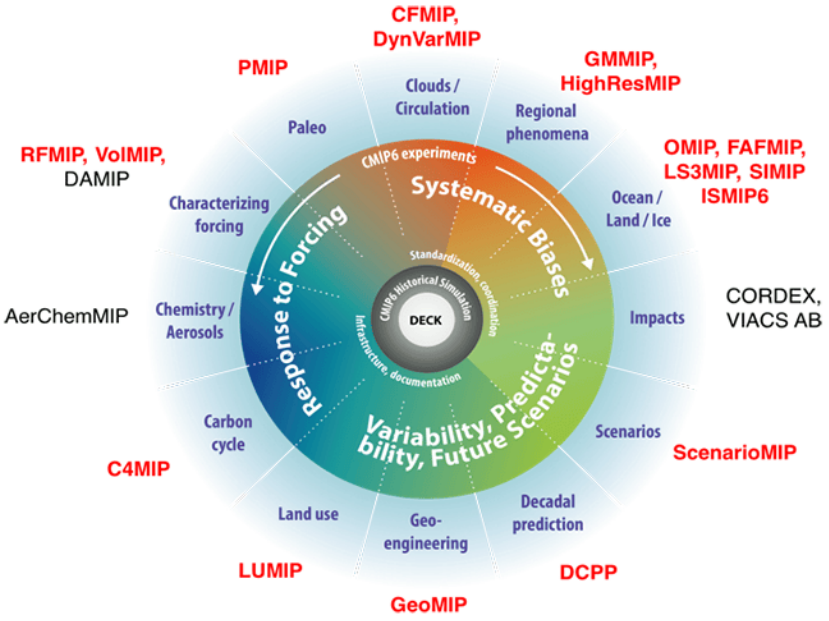


Figure 4.11: CMIP6 overview. *Figure 2 in Eyring et al. (2016).*

The MIPs focused on three main scientific questions:

1. How does the Earth system respond to forcing?
2. What are the origins of systematic model biases?
3. How can we assess future climate change given internal climate variability, climate predictability, and uncertainties in scenarios?

They also addressed seven WCRP Grand Science Challenges:

1. Connection between clouds, circulation, and Earth's climate sensitivity
2. Melting (land) ice and global consequences
3. Climate extremes
4. Changes in water availability
5. Regional sea level changes
6. Carbon feedbacks in the climate system
7. Near-term climate predictions

4.3.2 A family of Earth system models

CMIP6 consisted of **100 distinct climate models from 49 modelling groups** (Figure 4.12). Of these, ∼30 models from 21 modelling centres contributed results, and multiple models with shared components from a single modelling centre were often included, contributing to the clustering of model results.



Figure 4.12: CMIP6 model families. *From: NASA Goddard Institute.*

Exercise: Investigating Differences in Temperature and Precipitation between CMIP5 and CMIP6

This exercise will give you an opportunity to explore temperature (*tas*) and precipitation (*pr*)projections from the 1% yr^{-1} idealized emission scenarios from both CMIP5 and CMIP6. The aim will be to explore differences in both the timeseries, as well as the spread in *tas* and *pr* projections from the two generations of CMIP.
You can access the exercise here: https://github.com/mjma5t/ESM_clean/blob/main/Lab4/NSERC-CREATE-Module4-Lab-Updated.ipynb

4.4 CMIP6 — ScenarioMIP

4.4.1 What are Shared Socioeconomic Pathways?

SSP/RCP scenarios (Figure 4.13) are created by **social scientists** using Integrated Assessment Models (IAMs) — not by climate modellers. The scenarios map multiple plausible futures arising from differing assumptions about economic growth, population change, and technological evolution. They are then used by modelling centres in CMIP to drive future climate projections: models take the future concentrations of greenhouse gases, aerosols, and other forcing agents associated with each SSP, and simulate the evolution of the climate system in response.

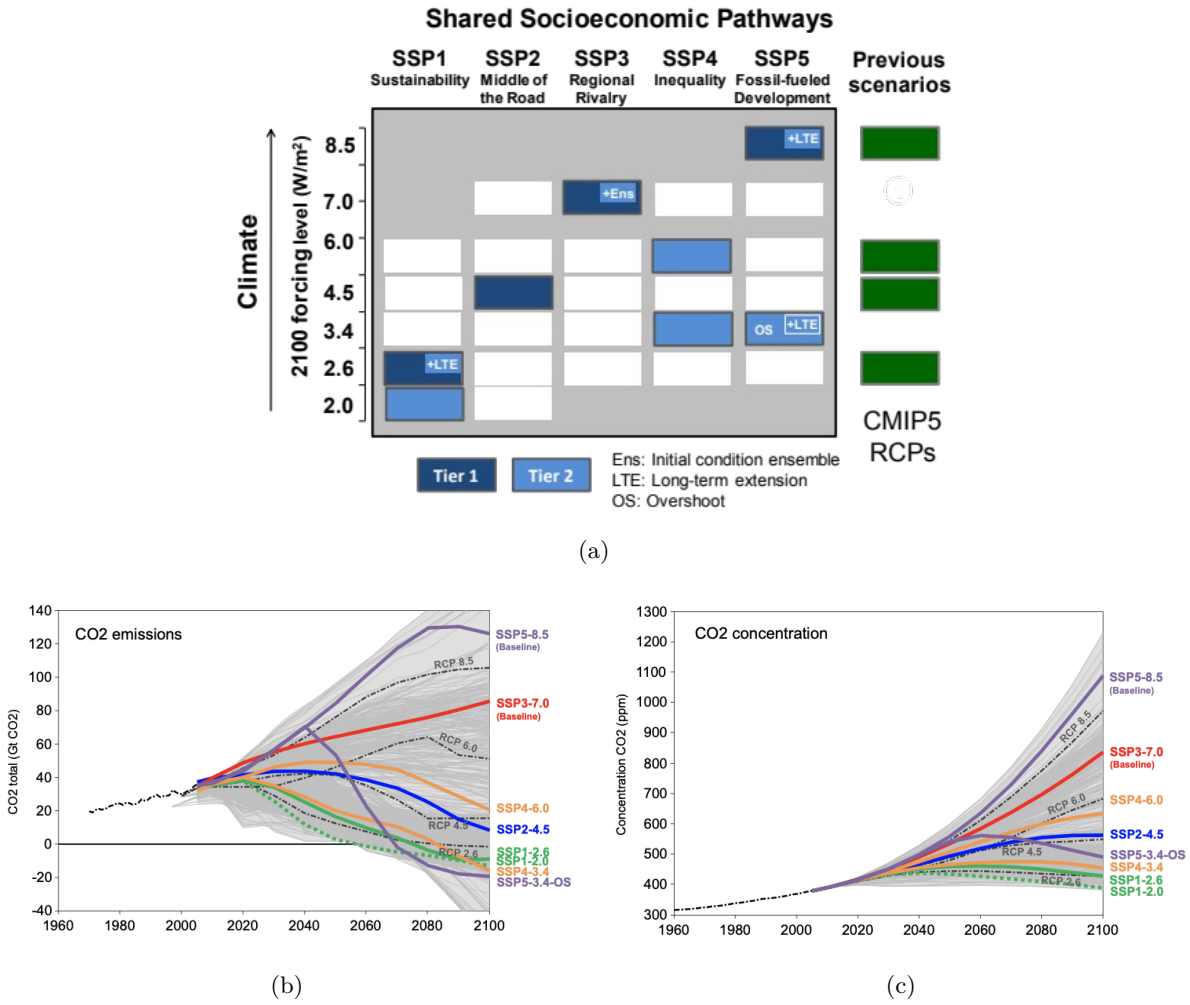


Figure 4.13: SSPs in CMIP6: *(a)* scenario matrix, *(b)* CO_2 emissions, *(c)* CO_2 concentration. *Figure 2 in O'Neill et al. (2016).*

4.4.2 ScenarioMIP results

Precipitation is the harder problem

Precipitation shows **larger uncertainty than temperature** (Figures 4.14 and 4.15), with a large proportion of it associated with internal variability until mid-century. Meanwhile for temperature, internal variability only accounts for a small fraction of the uncertainty, with structural uncertainty and scenario uncertainty accounting for the bulk of it. The partitioning of uncertainty in future climate climate projections will be explored in more detail in Module 6.

Variable	SSP1-1.9	SSP5-8.5
Warming, 2081–2100 vs. 1995–2014	0.69 °C	3.99 °C
Precipitation change	2.0 %	7.3 %

Table 4.3: CMIP6 multi-model ensemble mean change at the end of the 21st century, for the lowest and highest emissions scenarios (Tebaldi et al., 2021).

Table 4.3 summarizes the differences in global mean warming between the SSP1-1.9 and SSP5-8.5 scenarios, representing the lowest and highest emission scenarios from CMIP6. We see in Figure

4.15 that precipitation displays a relatively large inter-model spread (as represented by the 95% confidence interval) relative to temperature (Figure 4.14); a function of its larger uncertainty. Note that number of models included in each scenario is shown in the figure legends, ranging from 13 models for the SSP1-1.9 scenario to 38 models over the historical period, and for the SSP5-8.5 scenario.

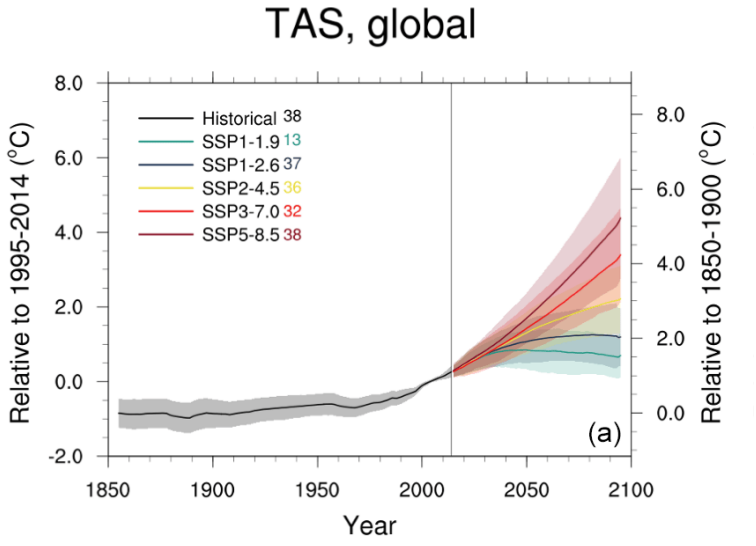


Figure 4.14: Global mean temperature projections. *Figure 1a in Tebaldi et al. (2021).*

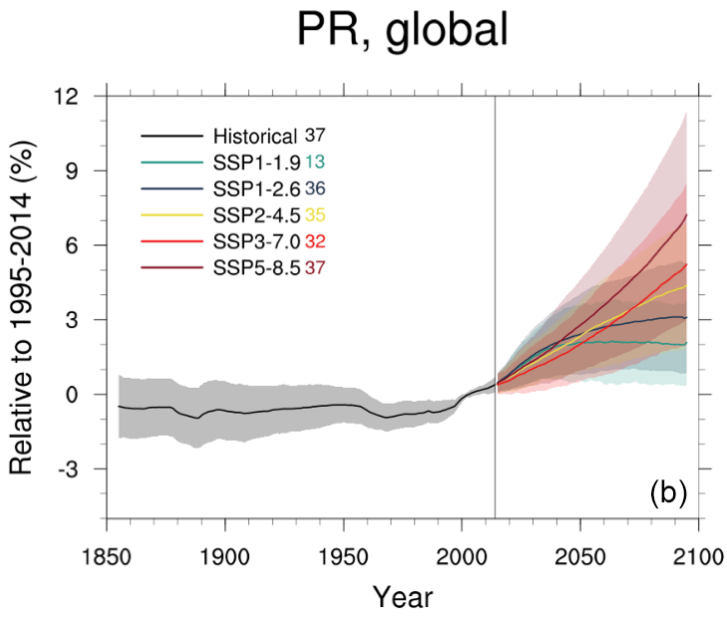


Figure 4.15: Global mean precipitation projections. *Figure 1b in Tebaldi et al. (2021).*

4.5 CMIP6 - Cloud Feedbacks Cause Higher Climate Sensitivity

The **CMIP6 mean equilibrium climate sensitivity (ECS)** and the **spread in ECS across CMIP6 models** is larger than for CMIP5 models:
The larger ECS in CMIP6 can primarily be linked to larger cloud feedbacks over the extratropics, relative to CMIP5 (Figure 4.16). This arises because low cloud cover and water vapour show relatively little sensitivity to sea surface temperatures in CMIP6, resulting in larger reductions in cloud cover, with comparatively smaller increases in cloud water vapour (Figure 4.17) (Zelinka

Metric	CMIP5	CMIP6
Mean ECS	3.3 K	3.9 K
Spread in ECS	2.1 – 4.7 K	1.8 – 5.6 K

Table 4.4: Comparison of the mean ECS and spread in ECS across CMIP5 and CMIP6 models (Zelinka et al., 2020).

et al., 2020).

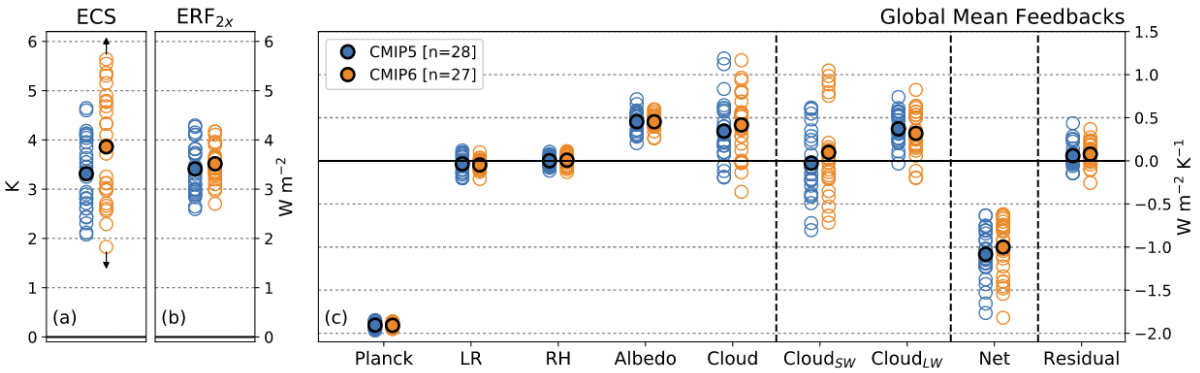


Figure 4.16: Global mean feedbacks. *Figure 1 in Zelinka et al. (2020).*

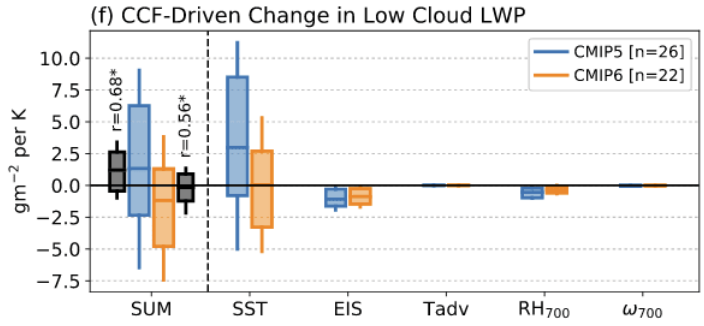


Figure 4.17: CCF-driven change in low cloud LWP. *Figure 3 in Zelinka et al. (2020).*

4.6 CMIP7 overview

The 7th Assessment Report and the CMIP7 intercomparison cycle are currently being planned (Figure 4.18). CMIP7 aims to address four fundamental research questions:

1. **Patterns of sea surface change:** how will tropical ocean temperature patterns co-evolve with those at higher latitudes?
2. **Changing weather:** how will dangerous weather patterns evolve?
3. **Water–carbon–climate nexus:** how will Earth respond to human efforts to manage the carbon cycle?
4. **Tipping points:** what are the risks of triggering irreversible changes across possible climate trajectories?

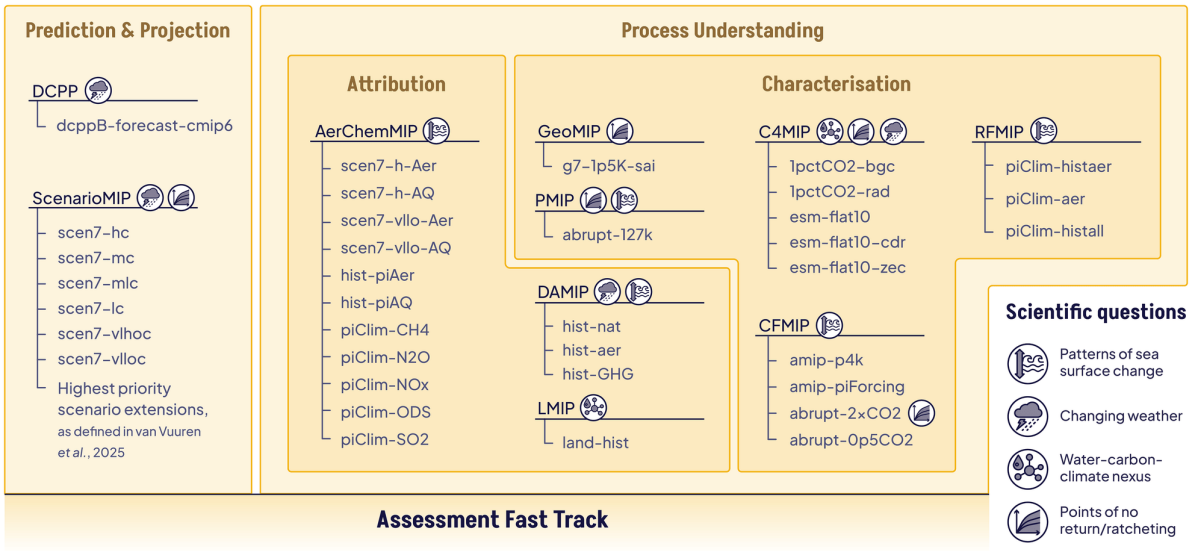


Figure 4.18: CMIP7 overview. *Figure 3 in Dunne et al. (2025).*

As with CMIP6, it will comprise a series of MIPs built on a base DECK experiment in which all modelling centres will be required to participate. The DECK experiment will include the following scenarios:

- Historical
- AMIP
- `piControl`
- `piClim-control` (model-specific ERF calculations)
- `piClim-anthro` (present-day total anthropogenic ERF)
- abrupt-4×CO_2
- 1 % yr^{-1} CO_2
- `piClim-4xCO2` (ERF of the 4×CO_2 scenario)

Two structural changes will distinguish CMIP7 from its predecessors, including an **Assessment Fast Track** incorporating extensive community input, intended to energise research driven by emergent advances and modelling centre priorities. **Community MIPs will be run on self-determined timelines**, instead of being limited to a common set of timelines, as they were in CMIP6 (Figure 4.19).

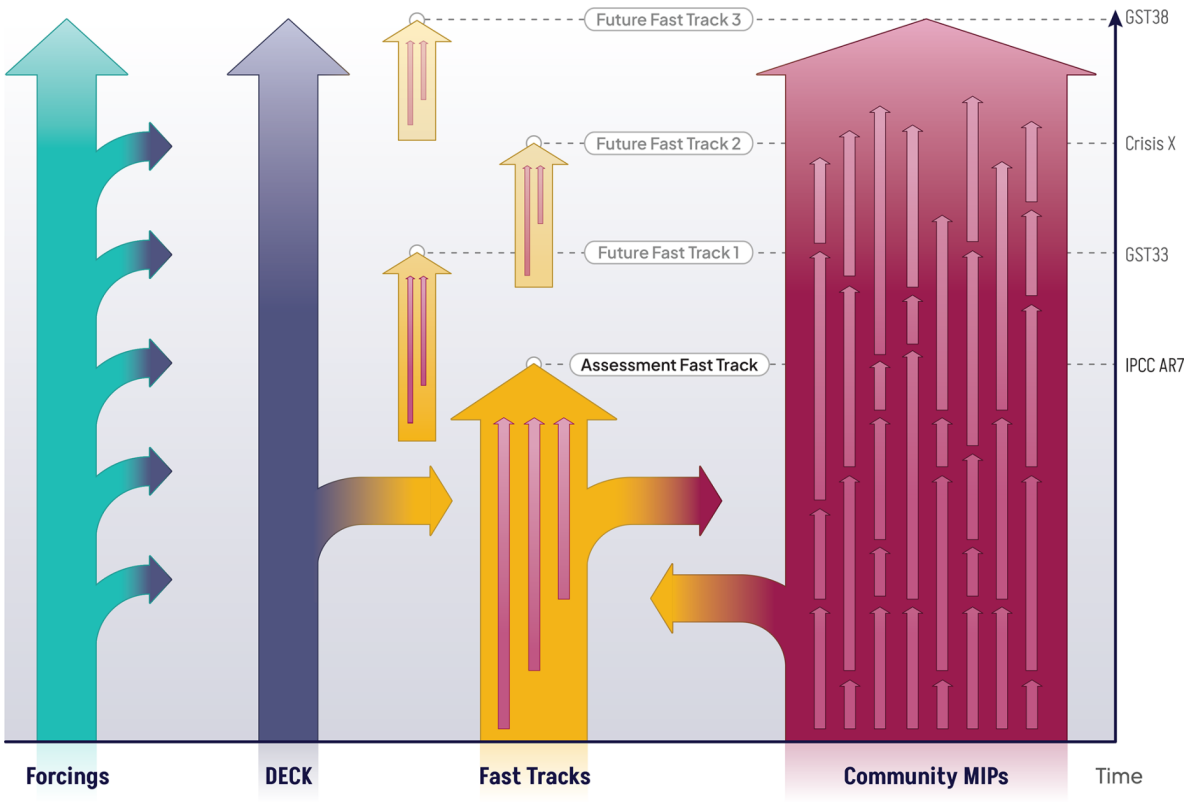


Figure 4.19: Schematic of the CMIP7 experiment design. *Figure 2 in Dunne et al. (2025).*

MODULE 5

Detection and Attribution

Learning outcomes

1. Understand the difference between detection and attribution in climate modelling.
2. Discuss the process of D&A in CMIP6, and understand the contributions to observed warming from different forcing agents.
3. Apply your understanding of D&A to extreme precipitation, wind, and snow cover extent.

5.1 Detection and attribution defined

The terms detection and attribution are often used together, but they answer different questions:

Detection of climate change

The process of demonstrating that climate, or a system affected by climate, **has changed in some defined statistical sense**, *without* providing a reason for that change (Bindoff et al., 2014). An identified change is detected in observations if its likelihood of occurrence by chance due to internal variability alone is determined to be small (Hegerl et al., 2010). In practice this means testing whether observed changes or variability fall within the bounds of natural internal variability. By the mid-1990s, global average warming over the preceding 50 years showed a detectable signal above and beyond natural variability (Stott and Mitchell, 2021).

Attribution of climate change

The process of **evaluating the relative contributions of multiple causal factors** to a change or event, with an assignment of statistical confidence (Bindoff et al., 2014). Attribution is more complex than detection, as it combines both statistical analysis with a physically based understanding (Hegerl and Zwiers, 2011). Critically, attribution **requires models**. It cannot be done from observations alone, because it requires a counterfactual scenario; what the climate would have done in the absence of a given forcing, and no such observation exists.

5.1.1 Process of Detection and Attribution (D&A)

There are four core elements to detection and attribution:

1. **Observations of one or more climate variables** (e.g. surface air temperature), that are understood, on physical grounds, to be relevant to the process in question
2. **An estimate of how external drivers of climate change have evolved** before and during the period under investigation, including both the driver whose influence is being investigated (such as rising GHG levels) and potential confounding drivers (such as solar activity)
3. A quantitative physically based understanding, normally encapsulated in a model, of **how these external drivers are thought to have affected these observed climate variables**

4. An estimate, often but not always derived from a physically based model, of **the characteristics of variability expected in these observed climate variables** due to random, quasi-periodic and chaotic fluctuations generated in the climate system that are not due to externally driven climate change

5.1.2 Approaches to D&A

Several methods have been used to detect an anthropogenic signal in the climate record:

- **Observation-based approaches** are difficult in practice because substantial assumptions are required, but include spatial pattern analysis (to separate variability from forcing), timeseries methods, and forcing history methods
- **Model-based comparisons**, the most common approach, compare observations against climate model data run with different combinations of forcing agents.

For attribution specifically, the literature uses filtering, fingerprinting, and regression approaches.

5.2 The basic D&A procedure

1. **Determine domain, period of interest, and filtering to be applied.**
   a. e.g. global, 1901–2010, spatial smoothing with decadal averaging.
2. **Gather data.**
   a. Observational data.
   b. Ensembles of historical climate runs — with natural (NAT) forcing only, and with ALL forcings.
   c. Control runs, to estimate internal variability.
3. **Process data.**
   a. *Observations:* homogenise, grid, identify missing values.
   b. *Historical model runs:* mask and collocate to match the available locations and dates/times of observations; process each individual run; ensemble-average to estimate signals.
   c. *Control runs:* divide into temporal chunks, then process similarly to the historical runs.

> **Why masking matters**
>
> Step 3(b) is easy to overlook and easy to get wrong. Model output is complete everywhere; observations are not. Comparing a globally complete model field against a spatially patchy observational record will produce a difference that reflects sampling, not physics. Collocation can reduce some of this uncertainty, though never fully remove it.

5.3 History of Detection and Attribution

Beginning with the IPCC's SAR (1995), there has been a chapter included on D&A in IPCC reports, with attribution statements on the role of greenhouse gases in accounting for observed warming trends (Figure 5.1). With each subsequent report, our confidence in attributing the warming to anthropogenic greenhouse gas emissions has increased, with the AR6 noting:

> it is **unequivocal** that human influence has warmed the global climate system since preindustrial times (IPCC, 2021).

5.4 CMIP6 DAMIP experiments

Unforced control simulations and historical simulations including the major anthropogenic and natural forcings are conducted as part of the DECK. **DAMIP** is the MIP focused specifically on

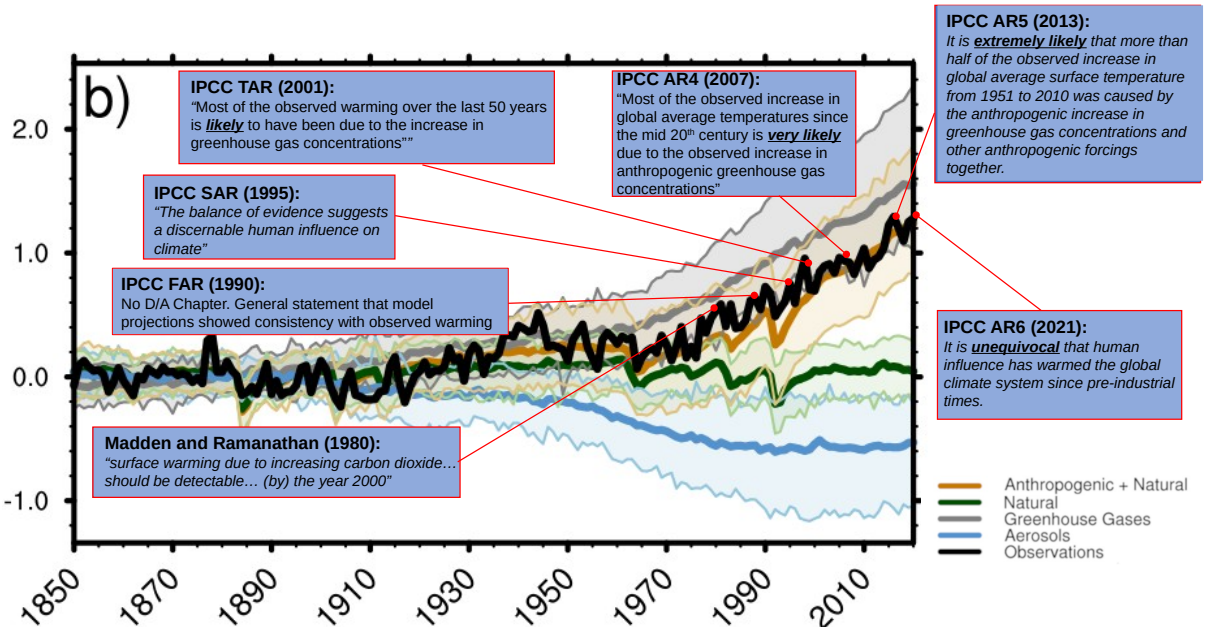


Figure 5.1: IPCC attribution statements set against the observed and DAMIP ensemble-mean temperature record. *Adapted from Figure 3.9b in Eyring et al. (2023).*

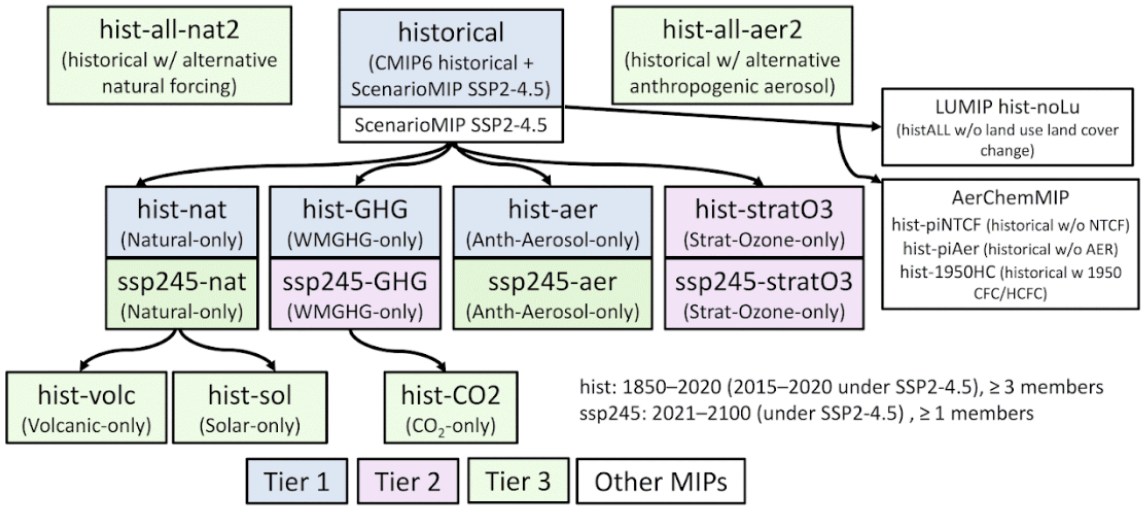


Figure 5.2: CMIP6 DAMIP experiments. *From: The Detection and Attribution Model Intercomparion Project.*

detection and attribution in CMIP6. It covers simulations over the historical period driven by **individual forcings** or subsets of forcings, and included 23 contributing models as of July 2023. The key novel features of the DAMIP experimental design are:

- new historical simulations of aerosols-only, stratospheric-ozone-only, CO_2-only, solar-only, and volcanic-only forcing, enabling improved estimation of the climate response to *individual* forcings;
- future single-forcing experiments, allowing observationally constrained projections of future climate change;
- an experimental design allowing models with and without coupled atmospheric chemistry to be compared on an equal footing.

Exercise: Detection and Attribution in CMIP6

This exercise will give you an opportunity to practice skills pertaining to the detection and attribution of climate change, by working with the HadCRUT5 temperature dataset, as well as output from the CMIP6 DAMIP experiments.
You can access the exercise here: https://github.com/mjma5t/ESM_clean/blob/main/Lab5/NSERC-CREATE-Module5-Lab-Updated.ipynb

5.5 CMIP6 DAMIP results

The observed air temperature trend from HadCRUT5 is shown as a black line, while the CMIP6 ensemble means from each DAMIP experiment are shown as different coloured lines. It is clear that the observed temperature timeseries is only able to be reproduced when all anthropogenic and natural forcings are included (Anthropogenic + Natural) (Figure 5.3).

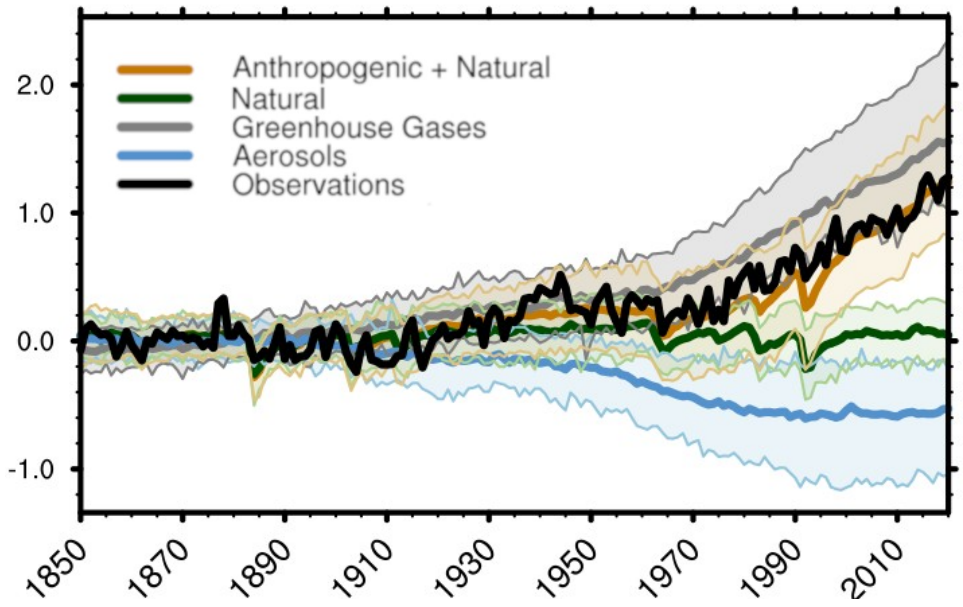


Figure 5.3: Observed and DAMIP ensemble-mean global surface temperature. *Figure 3.9b in Eyring et al. (2023).*

Greenhouse gases are the primary driver of the observed warming, while the net anthropogenic influence is offset by cooling from aerosol emissions. The net warming attributable to natural forcings (such as changes in solar radiation) and to internal variability, however is negligible, and centred around 0°C (Table 5.1).

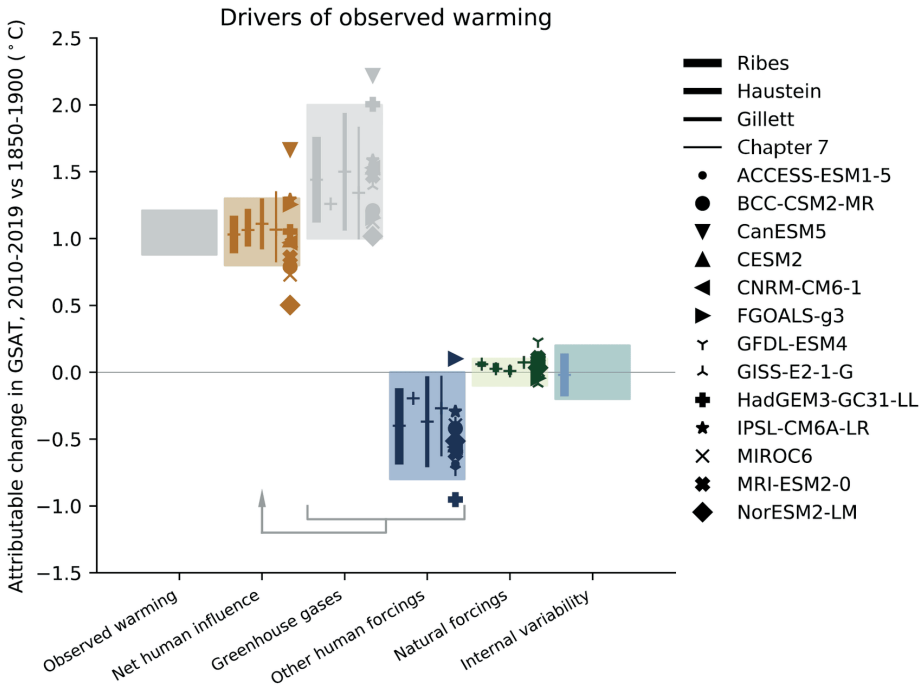


Figure 5.4: Drivers of observed warming. *Figure 3.8 in Eyring et al. (2023).*

Contribution to global mean surface air temperature	**Likely range**
Total anthropogenic	0.8 – 1.3 °C
Greenhouse gases	1.0 – 2.0 °C
Anthropogenic aerosols	−0.8 – 0.0 °C
Natural forcings	−0.1 – 0.1 °C
Internal variability	−0.2 – 0.2 °C

Table 5.1: Attributable contributions to the change in global mean surface air temperature over 2010–2019 relative to 1850–1900. Note that changes due to *specific* forcings are less certain than the totals for natural and anthropogenic forcing — the greenhouse gas and aerosol terms are large and partially offsetting.

5.6 D&A of extreme precipitation: the Asian monsoon

Dong et al. (2022) showed that between 1950–2014, simulated annual maximum 5-day (Rx5day) and 1-day (Rx1day) precipitation over Asia showed opposing responses to anthropogenic emissions greenhouse gases and aerosols (Table 5.2). They note three main findings:

- The observed increases are **attributable to anthropogenic greenhouse gas emissions**.
- Aerosols have likely had an **offsetting effect** on the observed extreme precipitation trends; which is why the observed Rx5day trend is smaller than the GHG-only response.
- Signals from natural forcings (panels i, j) were **not detectable** above internal variability.

5.7 D&A of mean and extreme near-surface winds

Chou et al. (2025) compared trends over 1980–2019 in extratropical low-level mean and extreme (>90th percentile) winds in reanalyses and CMIP6 models. For the Southern Hemisphere, both mean and extreme winds increased by$\sim$0.1 m s^{-1} decade^{-1} in reanalyses (Figure 5.6). It was found that:

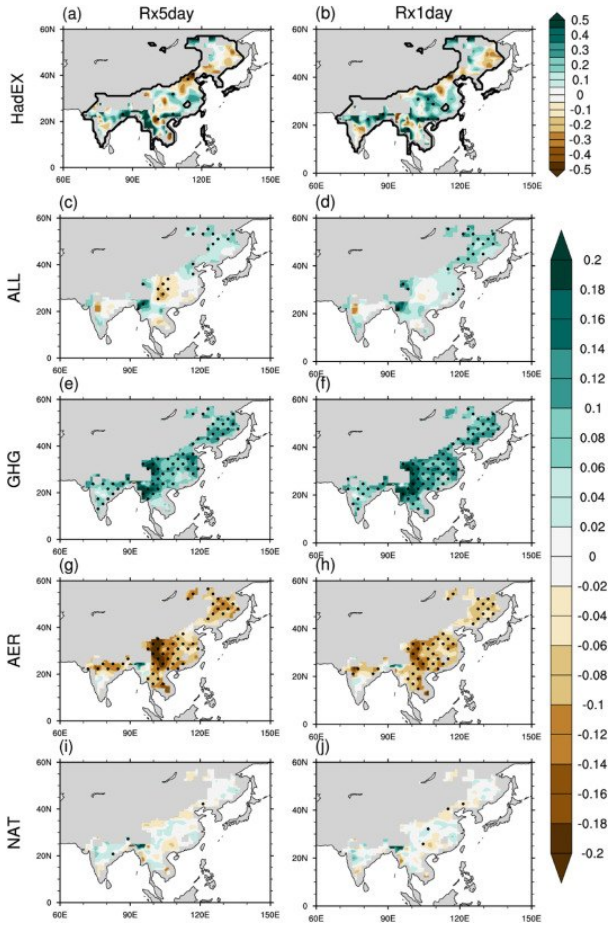


Figure 5.5: Trends in extreme precipitation over Asia. *Figure 1 in Dong et al. (2022).*

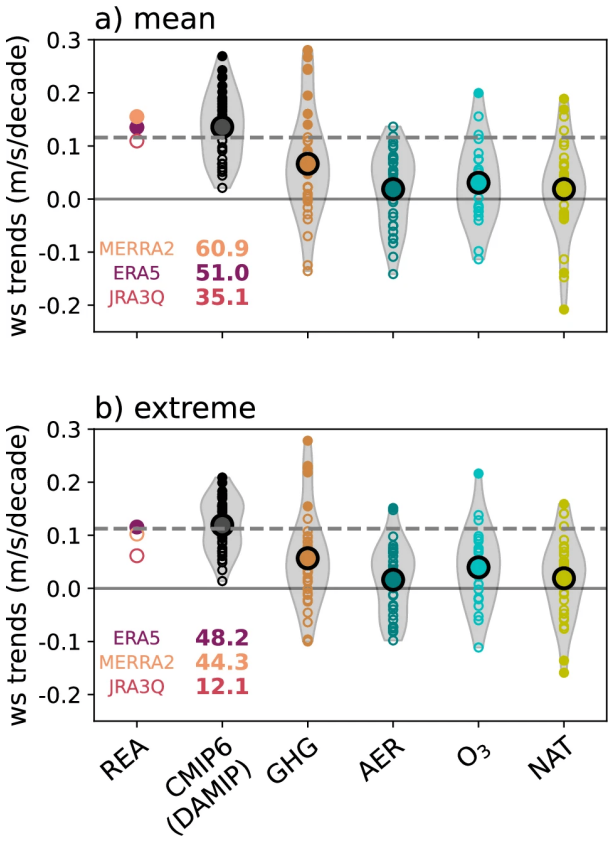


Figure 5.6: Attribution of human influence on austral wind speeds. *Figure 2 in Chou et al. (2025).*

Forcing / source	Rx5day	Rx1day
Anthropogenic GHG (panels e, f)	9.70 % century^{-1}	10.86 % century^{-1}
Aerosols alone (panels g, h)	−8.99 % century^{-1}	−7.01 % century^{-1}
Observed, HadEX (panels a, b)	6.04 % century^{-1}	10.96 % century^{-1}

Table 5.2: Simulated and observed trends in annual maximum 5-day and 1-day precipitation over the Asian monsoon region, 1950–2014.

- CMIP6 closely reproduce the observed increase **only when all forcings are included**.
- The increase is mostly attributable to **greenhouse gas and stratospheric ozone** forcings.

5.8 D&A of Northern Hemisphere snow cover extent

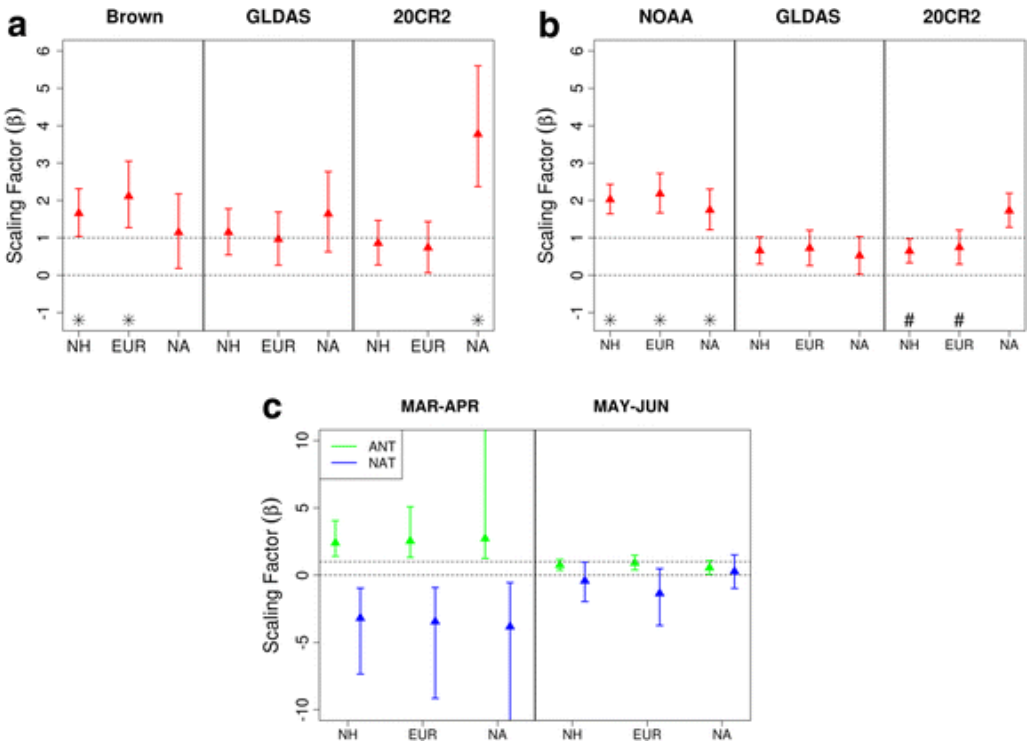


Figure 5.7: Attribution of human influence on NH snow cover extent. *Figure 4 in Najafi et al. (2016).*

Najafi et al. (2016) examined the contributions of natural and anthropogenic forcings to changes in NH snow cover extent (SCE) over North America and Eurasia (Figure 5.7). They note that the anthropogenic signal was most detectable over March and April (panels a and c):

- There is **considerable spread between observational estimates**, which is itself part of the problem, as this indicates substantial uncertainty.
- The Brown dataset shows an observed decline in March/April SCE of $\sim 0.8 \times 10^6$ km^2 decade^{-1} since 1970.
- The 1968–2012 decline in May–June SCE is estimated at 0.94×10^6 km^2 decade^{-1}.

The results show that the observed decline is **inconsistent with internal variability or natural forcings alone**:

- The largest anthropogenic signal is over North America, and in March/April, driving the

observed declines.

- Natural forcings alone would have contributed to an *increase* in snow cover, thus the anthropogenic signal must be larger than the raw observed trend.

Land model choice is a limiting factor

SCE shows substantially greater uncertainty than temperature, and the simulations are sensitive to the choice of land model: roughly 30 % of the scaling factor variance can be explained by land model biases. This is an example of structural uncertainty (Module 4).

MODULE 6

Quantifying Model Uncertainty

Learning outcomes

1. Understand how uncertainty in future climate scenarios is calculated and partitioned into its constituent components — internal variability, model uncertainty, and scenario uncertainty.
2. Appreciate how the contributions of each component vary across temporal and spatial scales, and how they differ between precipitation and temperature.
3. Apply a Hawkins and Sutton (2009, 2011) type uncertainty analysis to CMIP6 model output.

6.1 Sources of uncertainty

This module brings together the four categories introduced in Module 4. Future climate projections are typically considered to carry four types of uncertainty:

- **Internal variability**, assessed from the variability of the climate over the historical period (Module S1);
- **Structural uncertainty**, from differences in model physics and resolution;
- **Parametric uncertainty**, from poorly constrained parameters (Module S2);
- **Scenario uncertainty**, seen as the spread between the ensemble means of the different scenarios.

Model uncertainty

Model uncertainty (Figure 6.1) is the combination of structural and parametric uncertainty, and represents uncertainties arising due to differences in model structure, resolution, and parameterization. In the Hawkins and Sutton (2009, 2011) framework, model uncertainty is represented by the spread in model projections *within a single emissions scenario*. Note that this approach does **not separate model uncertainty into its structural and parametric parts**; they are considered together.

6.2 Quantifying uncertainty

Total uncertainty, following Hawkins and Sutton (2009), is assumed to be the linear combination of internal variability I, model uncertainty $M(t)$, and scenario uncertainty $S(t)$:

$$T(t) = I + M(t) + S(t). \tag{6.1}$$

Note that I carries no time dependence in this framework, while M and S do; leading to increasingly large contributions from model and scenario uncertainty over the course of climate projections.

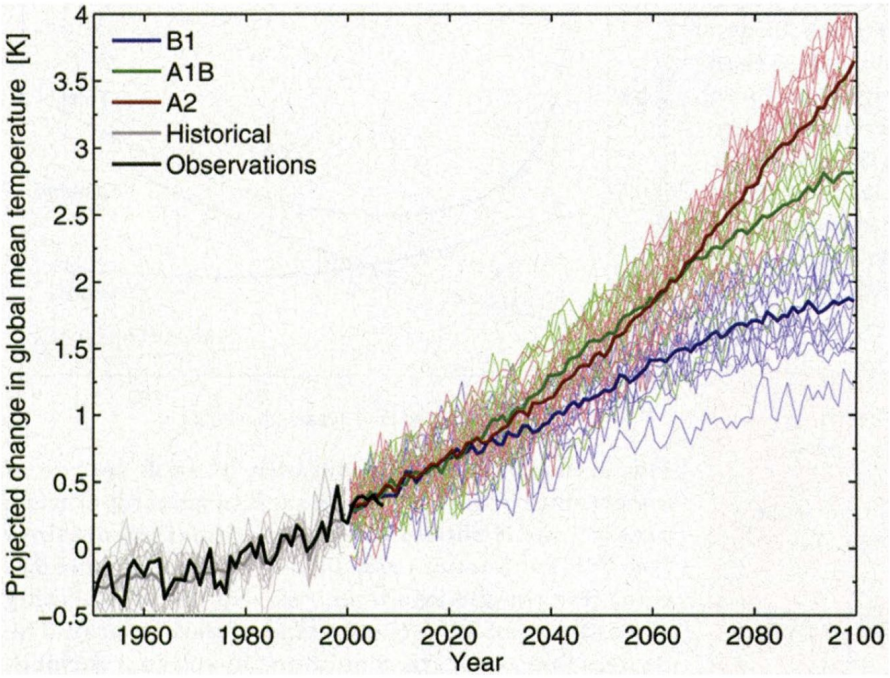


Figure 6.1: Sources of uncertainty in a multi-model ensemble. *Figure 1 in Hawkins and Sutton (2009).*

6.2.1 Assessing internal variability

Following Hawkins and Sutton (2009, 2011): for each model and scenario, fit a 4th order polynomial to the timeseries. Each projection is then modelled as

$$X_{m,s,t} = x_{m,s,t} + i_{m,s} + \varepsilon_{m,s,t}. \tag{6.2}$$

where

$X_{m,s,t}$ the projection, for model m, scenario s, time t
$x_{m,s,t}$ the smoothed polynomial fit
$i_{m,s}$ reference temperature, relative to a given year or years
$\varepsilon_{m,s,t}$ the residual, assumed to be internal variability

Models are then weighted by their ability to represent global mean warming over the historical period:

$$w_m = \frac{1}{x_{\text{obs}} + |x_{m,\text{ref}} - x_{\text{obs}}|}, \tag{6.3}$$

$$W_m = \frac{w_m}{\sum_m w_m}. \tag{6.4}$$

where

w_m raw weight for model m; larger when the model is closer to observations
W_m normalised weight, summing to unity across models
x_{obs} observed reference value
$x_{m,\text{ref}}$ model m's value over the reference period

Multi-model mean internal variability is the weighted mean of the individual model internal variabilities:

$$I = \sum_m W_m \, \sigma^2_{s,t}(\varepsilon_{m,s,t}). \tag{6.5}$$

A known underestimate

Equation (6.5) likely **underestimates** the true contribution of internal variability, because it is based on only one realization per scenario per model; unlike a SMILE, which explores a range of realizations per scenario for a single model. Lehner et al. (2020) discuss the advantages of using SMILEs to assess internal variability (see Section 6.4).

6.2.2 Estimating model uncertainty

Model uncertainty $M(t)$ is assumed to be the weighted variance of the individual fitted polynomial curves $x_{m,s,t}$:

$$M(t) = \frac{1}{N_s} \sum_s \sigma^{2,W}_m (x_{m,s,t}). \tag{6.6}$$

where

N_s number of scenarios
$\sigma^{2,W}_m$ weighted variance taken across models

6.2.3 Estimating scenario uncertainty

Scenario uncertainty $S(t)$ is estimated from the variance of the weighted multi-model means for each scenario:

$$S(t) = \sigma^2_s \left(\sum_m W_m \, x_{m,s,t} \right). \tag{6.7}$$

6.2.4 Fractional uncertainty

Calculate the mean change of all projections above the reference temperature, $G(t)$:

$$G(t) = \frac{1}{N_s} \sum_{m,s} W_m \, x_{m,s,t}. \tag{6.8}$$

The fractional uncertainty, for a given confidence level, is then

$$F(t) = \frac{z^* \sqrt{T(t)}}{G(t)}. \tag{6.9}$$

where

z^* critical value for the chosen confidence level
$T(t)$ total uncertainty from Equation (6.1)
$G(t)$ mean projected change above the reference

Expressing uncertainty as a *fraction* of the projected signal allows us to compare the relative contributions of different types of uncertainty under different climate scenarios, as well as across different spatial scales, and for different climate variables.

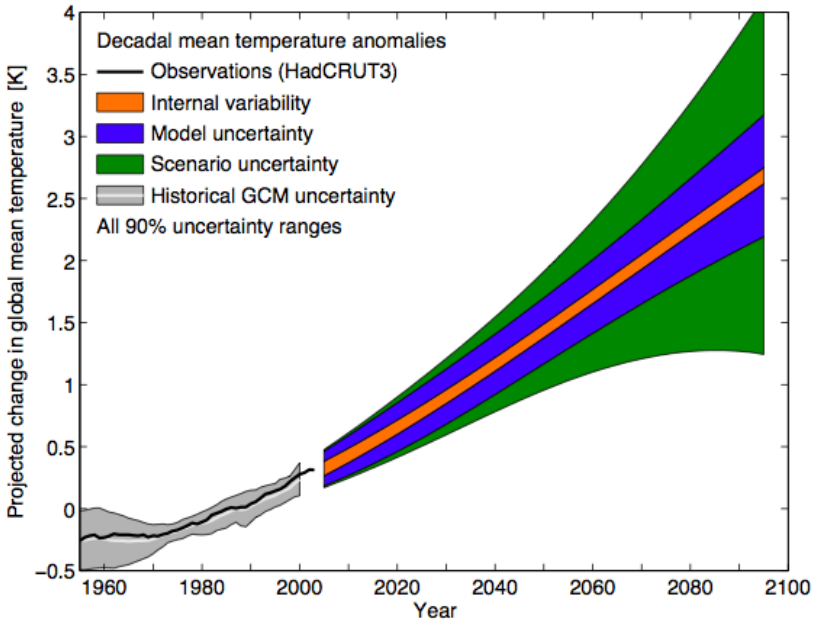


Figure 6.2: Relative contributions to uncertainty in global mean temperature projections. *Figure 2a from Hawkins and Sutton (2011).*

6.3 Sources of uncertainty: global means

In the near term, model uncertainty and internal variability account for the bulk of the uncertainty in global mean climate projections, whereas the contribution from scenario uncertainty becomes the most important as we approach the end of the 21st century.

Relative to global mean temperature, global mean precipitation projections show a larger contribution from internal variability. Model uncertainty is also larger than it is for temperature, and remains the dominant source of uncertainty throughout the 21st century.

	Precipitation	**Temperature**
Near term	Internal variability is a substantial fraction	Internal variability and model uncertainty dominate
Mid-century onward	Model uncertainty is largest *throughout*	Scenario uncertainty quickly becomes dominant
End of century	Scenario uncertainty grows in importance	Scenario uncertainty dominant

Table 6.1: How the uncertainty budget evolves for the two variables. The key asymmetry: for temperature, scenario uncertainty takes over from mid-century; for precipitation, *model* uncertainty remains the largest term throughout the century.

Why the asymmetry matters

For temperature, the dominant uncertainty by late century is **a choice we make** — which emissions scenario the world follows. For precipitation, it is **a limitation of our models**, and no emissions decision resolves it. The two variables therefore call for different responses: policy in one case, model development in the other.

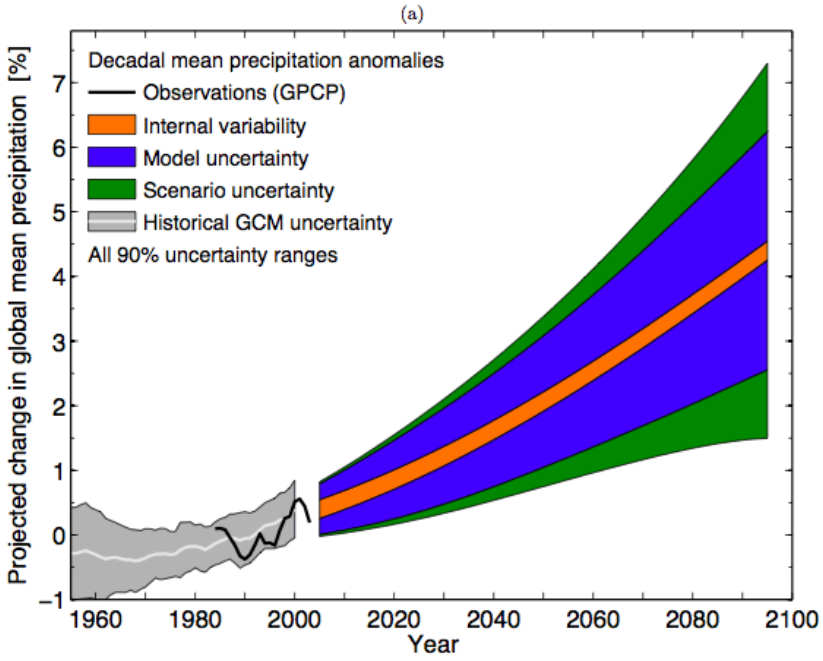


Figure 6.3: Relative contributions to uncertainty in global mean precipitation projections. *Figure 2b from Hawkins and Sutton (2011).*

6.4 Uncertainty in regional climate projections

Lehner et al. (2020) compare the contributions of internal variability to overall uncertainty in *global* projections of temperature and precipitation against *regional* case studies over the British Isles and the Sahel. Unlike Hawkins and Sutton (2009, 2011) who estimate internal variability from a single model realization, Lehner et al. (2020) estimate it using initial condition large ensembles (SMILEs):

- They find that the Hawkins and Sutton (2009, 2011) approach **underestimates** internal variability relative to a SMILE-based approach, particularly at smaller spatial scales; confirming the caveat noted after Equation (6.5).
- In agreement with Hawkins and Sutton (2011), for *global* mean projections internal variability is most important in the near term (2020s and 2030s), and its contribution to overall uncertainty is relatively small later in the century.
- For *regional* projections, however, internal variability remains an important source of uncertainty **throughout** the 21st century, particularly for precipitation.

Closing the loop

The reader began with the claim that climate is projectable even though weather is not, because forced signals emerge from internal variability given enough time and space to average over. This last result is the quantitative statement of the limits of that claim: shrink the spatial scale far enough, and internal variability never becomes negligible — no matter how far into the future you look, and no matter how good the model.

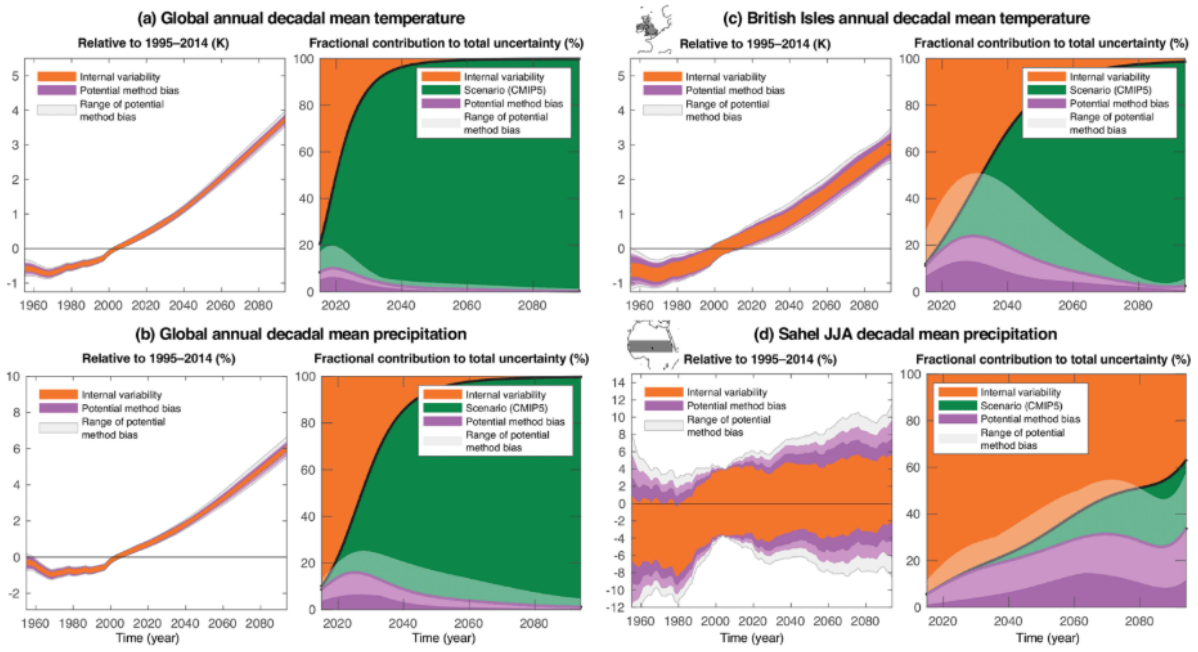


Figure 6.4: Uncertainty in global mean versus regional temperature and precipitation projections. *Figure 5 in Lehner et al. (2020).*

Supplementary Material

MODULE S1

Initial Condition Ensembles

Learning outcomes

1. Understand the connections between chaos theory, Lorenz attractors, and initial condition ensembles.
2. Explain how single model initial-condition large ensembles (SMILEs) are used to estimate internal variability.
3. Discuss the relative contributions of internal variability to temperature, precipitation, and snow cover extent.

S1.1 Initial condition ensembles

Initial condition ensembles (Figure S1.1) have their origins in numerical weather prediction:

- Early forecasts at ECMWF were based on a **single** forecast from one set of initial conditions.
- By the 1980s ECMWF began producing an *ensemble* of forecasts, each starting from slightly different initial conditions, in order to quantify forecast uncertainty. The approach was widely adopted through the 1990s.
- Forecast uncertainty is then calculated from the **spread** across the ensemble

For more information, readers are referred to a background primer from the Royal Meteorological Society on ensembles in numerical weather prediction: https://www.rmets.org/metmatters/ensembles-how-forecast-comes-together

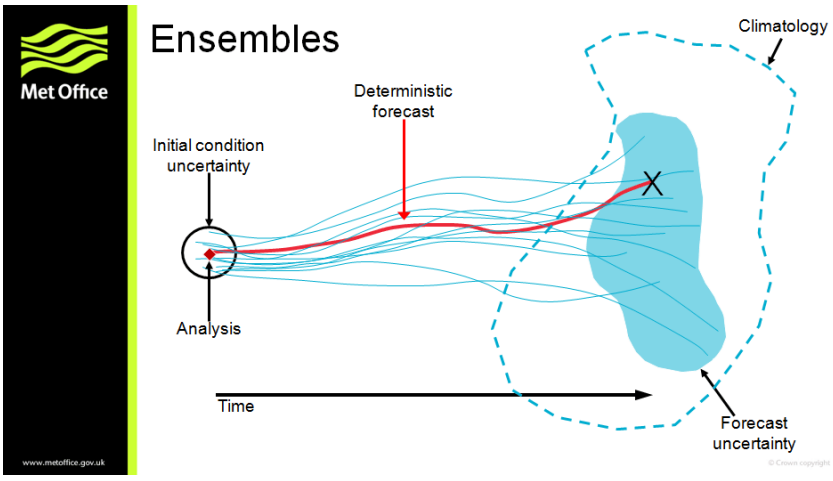


Figure S1.1: A conceptual diagram showing how initial condition ensembles are used in numerical weather prediction. *From the Royal Meteorological Society.*

Internal variability

Natural variability in the climate system occurring in the *absence* of external forcing, owing to processes intrinsic to the atmosphere, ocean, land, and cryosphere, and to the interactions between them.
The key implication: a single projection represents **one realization among many** that could have arisen within today's climate.

In climate science, internal variability is investigated using a large ensemble of runs from a single model: a **Single Model Initial Condition Large Ensemble**, or SMILE. SMILEs use **identical external forcing** across members, with only a slight difference in initial condition. Any spread between members is therefore internal variability by construction. SMILEs allow us to estimate the proportion of total uncertainty attributable to internal variability, and are also useful for exploring extreme events within a given climatology, since a large ensemble samples the tails far better than a single realization can.

S1.2 Lorenz attractors and chaos theory

Edward Lorenz was a meteorologist and mathematician who founded the field of chaos theory and applied it to numerical weather prediction. In the early 1960s, while working with an early numerical weather prediction model, Lorenz discovered that small rounding errors: **on the order of** 1×10^{-3}, carried over from a previous time step, could yield a drastically different forecast by the end of the simulation. To explore this, he constructed a set of three nonlinear differential equations describing the evolution of convection in a simplified model:

$$\frac{dx}{dt} = \sigma\,(y - x), \tag{S1.1}$$

$$\frac{dy}{dt} = x\,(\rho - z) - y, \tag{S1.2}$$

$$\frac{dz}{dt} = xy - \beta z. \tag{S1.3}$$

where

- x convective overturning rate
- y horizontal temperature difference
- z vertical temperature difference
- σ Prandtl number
- ρ Rayleigh number
- β geometric factor

Using parameter values $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$, Lorenz found that:

- the system produces a set of solutions that split into two main spiral branches — resembling butterfly wings — called **attractors** (Figure S1.2);
- solutions appear to jump randomly between the two branches, which is what we call **chaos**;
- while the solutions stayed within certain **bounds**, they were never repeated for different initial conditions.

Why this is the whole argument in miniature

The Lorenz system is bounded but unpredictable — exactly the property that makes climate projectable while weather is not. You cannot say which wing the trajectory sits

on at a given moment (weather), but you can characterise the attractor it is confined to (climate). Change ρ and the attractor itself changes shape: that is forcing.

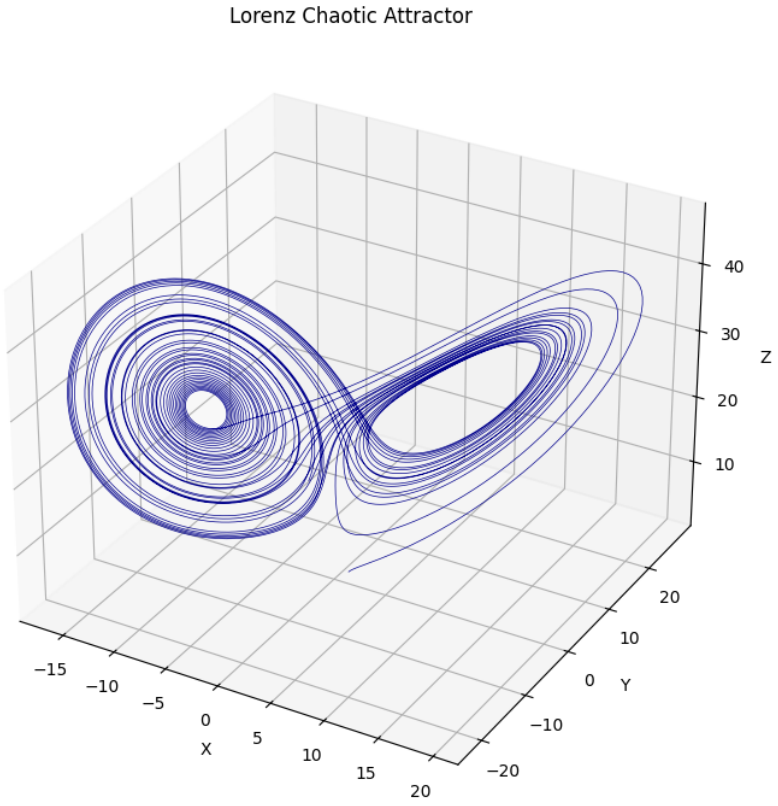


Figure S1.2: The numerical solutions of a Lorenz system of equations. *Used under a CC0 license from Lorepenoten.*

S1.3 Using large ensembles to explore internal variability

The first large ensemble was the Community Earth System Model (CESM) Large Ensemble (LE) (Kay et al., 2015). Another early example was the Canadian Earth System Model v2 (CanESM2) LE (Kirchmeier-Young et al., 2017; Kushner et al., 2018) (Figure S1.3). A large ensemble also exists for the Canadian Earth System Model v5 (CanESM5): the CanESM5 LE (Fyfe et al., 2021; Swart et al., 2019).

The total response within each ensemble member of a large ensemble is assumed to be a linear sum of the forced response and internal variability:

$$\text{TOTAL}_i = \langle \text{Forced} \rangle + \text{Internal}_i. \tag{S1.4}$$

where

TOTAL_i total response in ensemble member i
$\langle \text{Forced} \rangle$ forced response, common to all members
Internal_i internal variability in member i

The two terms are estimated as follows:

- The **forced response** is derived from the ensemble mean of the individual members. This assumes that averaging across a large number of members drives the contribution of internal variability toward zero.
- **Internal variability** is then the residual left over after the forced response is subtracted from each individual model run.

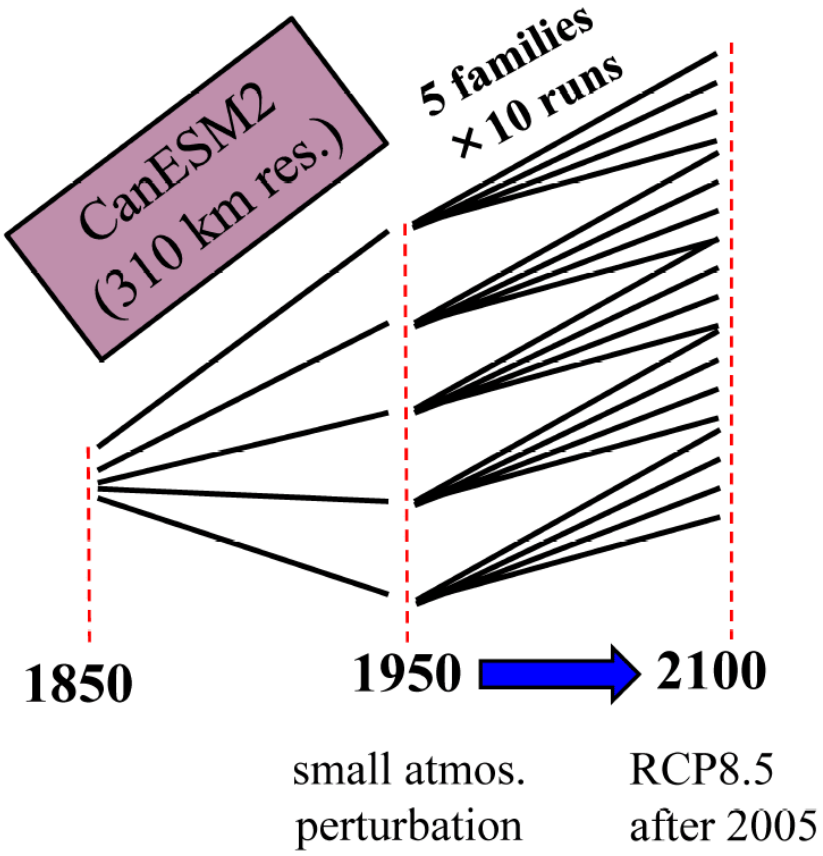


Figure S1.3: Conceptual diagram showing how the CanESM2-LE was constructed. *Figure 1 in Leduc et al. (2019).*

S1.4 Internal variability in temperature

Deser et al. (2020) showed that both the CESM LE and the MPI LE contain the average observed temperature trend over North America, yet their ensemble *means* are very different (Figure S1.4). The spread of each individual LE is similar in magnitude to the spread of the entire CMIP5 ensemble. Three implications follow:

- At regional scales it becomes **much harder to separate** the forced response from internal variability.
- The spread across the range of CMIP LE experiments is larger than the CMIP5 ensemble spread, suggesting that **CMIP5 underestimates** internal variability in regional temperature responses.
- While the number of LEs increased in CMIP6 relative to CMIP5, many models still have no LE experiments available, making their forced response difficult to determine — and making structural uncertainty hard to distinguish from internal variability.

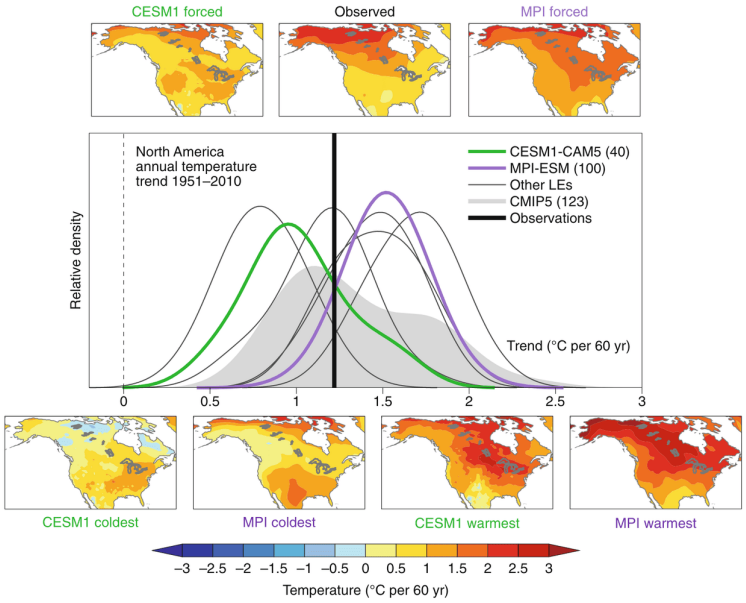


Fig. 1 | Internal variability and model differences in continental temperature trends. The distribution of 60-yr annual temperature trends (1951–2010) over North America (24–72° N, 180–62° W) from seven ESM LEs (thin curves), 40 different CMIP5 models (grey shading), and observations (Berkeley Earth Surface Temperature; vertical black line). The maps show the associated patterns of temperature trends: top row, observed and the forced component (estimated by the ensemble mean) from two LEs (CESM1 in green and MPI in purple); bottom row, individual ensemble members from CESM1 (green) and MPI (purple) with the weakest ('coldest') and strongest ('warmest') trends. Note that the individual member maps show the total (forced-plus-internal) trends in the model LEs. Observed trends are analogous to an individual ensemble member in that they reflect forced and internal contributions.

Figure S1.4: Internal variability and differences in the distribution of continental temperature trends between the CESM-LE and the MPI-LE. *Figure 1 in Deser et al. (2020).*

S1.4.1 Case study: the "hiatus"

In the 2010s, the question of why an apparent **hiatus** in warming had occurred was of enormous interest. The hiatus appears to have been largely an **artifact of a shift in observational records**, with an increasing contribution of data from ocean buoys, which record slightly cooler temperatures than ship-based measurements. After correcting for this change in data sources, the bias-corrected record no longer shows an apparent hiatus in the global mean (Figure S1.5). It nonetheless illustrates the importance of internal variability in accounting for short-term climate trends.

S1.4.2 Trend length and the role of internal variability

Kay et al. (2015) explored the variability of air temperature trends in the CESM LE, looking at both 10-year and 20-year trends (Figure S1.6). They showed that the spread in **10-year** temperature trends is much larger than for **20-year** trends; illustrating how internal variability dominates short-term trends. Trends over three periods were compared: trends in the 1850-control simulation, trends between 1990-2009, and trends between 2030-2049. The main findings were:

- In a pre-industrial climate, **roughly equal proportions of ensemble members show warming and cooling trends**, with a mean centred on zero, as expected in the absence of external forcing. The variability seen is therefore due to internal variability alone.
- The confidence intervals of both the CESM-LE and the observed estimates of the 1990–2009 10-year trend **contain zero warming**, and even slight (statistically insignificant) cooling. Even in periods where the long-term trend is significantly positive, internal variability alone can generate short intervals of little to no warming; particularly at regional scales.
- By 2030–2049, **none** of the individual LE members show negative 10-year trends, indicating

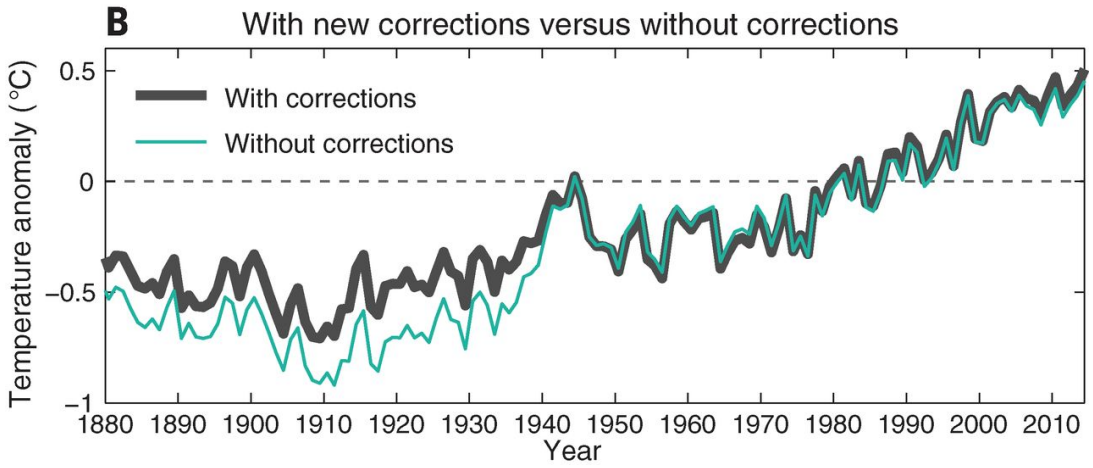


Figure S1.5: Global (land plus ocean) surface temperature anomaly time series with and without time-dependent bias corrections. *Figure 2b in Karl et al. (2015).*

that the forced response has begun to outweigh internal variability.

S1.5 Internal variability in precipitation

Deser et al. (2014) estimated the contributions of the forced response and internal variability to (a) winter and (b) summer precipitation trends between 2000 and 2060 over North America in the CCSM3 LE (Figure S1.7). They found that:

- The forced response is **relatively small** compared with the contribution of internal variability, in both winter and summer.
- The variability is large enough that individual ensemble members show a wide range (and sometimes opposing precipitation patterns); as shown in the sub-panels of the figure.

S1.6 Internal variability in snow cover extent

Kushner et al. (2018) explored the spread in estimates of 1981–2005 Canadian snow cover climatologies and trends in the CanESM2 initial condition LE, comparing the results against the CCSM4 and CESM LE experiments (Figure S1.8). The spread in ensemble *means* across the three large ensembles is of similar magnitude to the spread of the individual LEs, while:

- Internal variability accounts for a **substantial amount of the uncertainty** in the observed 1981–2005 snow cover extent trend.
- CanESM2 exhibits a noticeably larger decline in snow cover than the other models. This is a consequence of the fact that **CanESM2's ECS sits at the higher end of the CMIP5 estimates**, and it therefore exhibits a larger global mean warming over the historical period than CCSM4 or CESM.

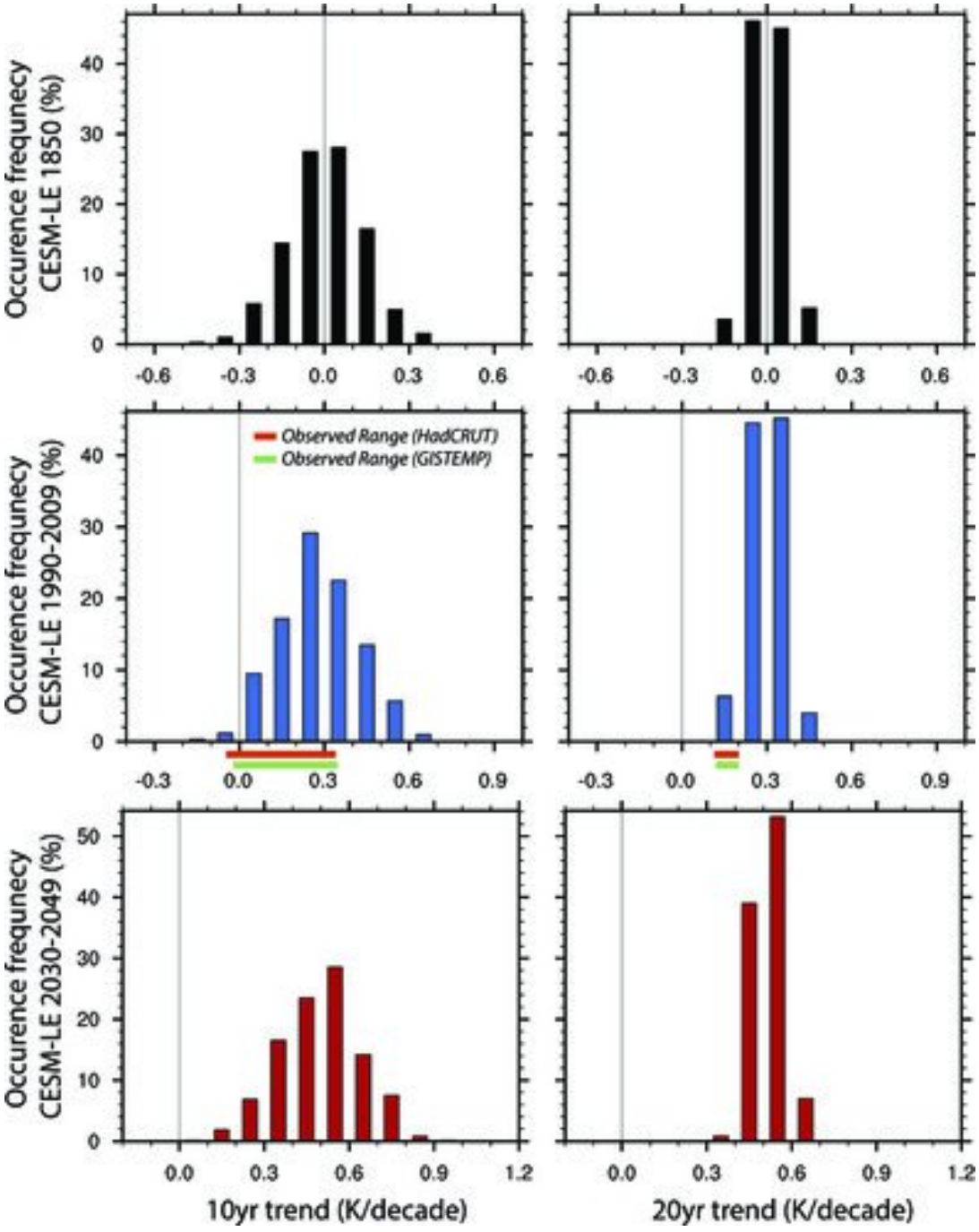


Figure S1.6: Histograms of 10-year and 20-year trends in global surface air temperature for the 1850 control (black), starting from 1990 to 2009 (blue) and starting from 2030 to 2049 (red). *Figure 3 in Kay et al. (2015).*

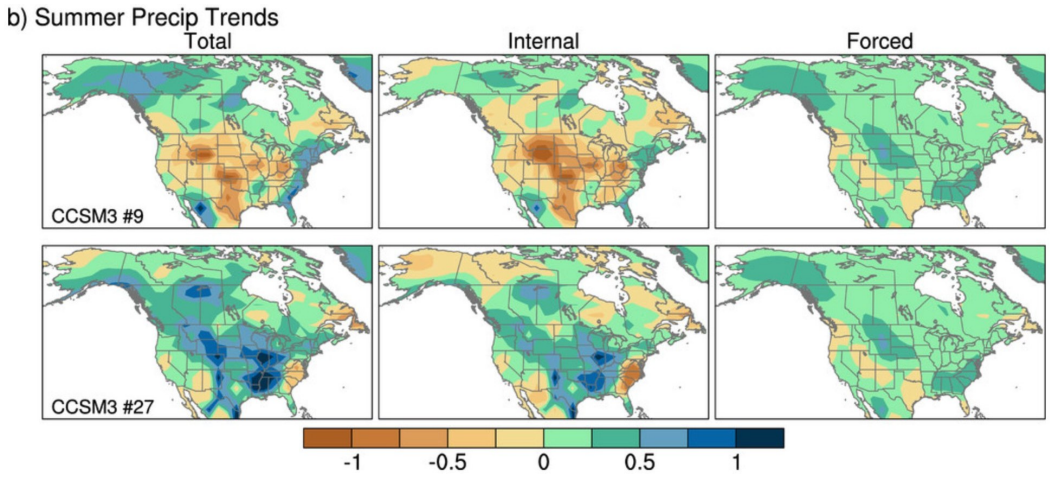


Figure S1.7: Total 2010–60 summer precipitation trends decomposed into (center) internal and (right) forced components for two contrasting CCSM3 ensemble members. *Figure 3b in Deser et al. (2014).*

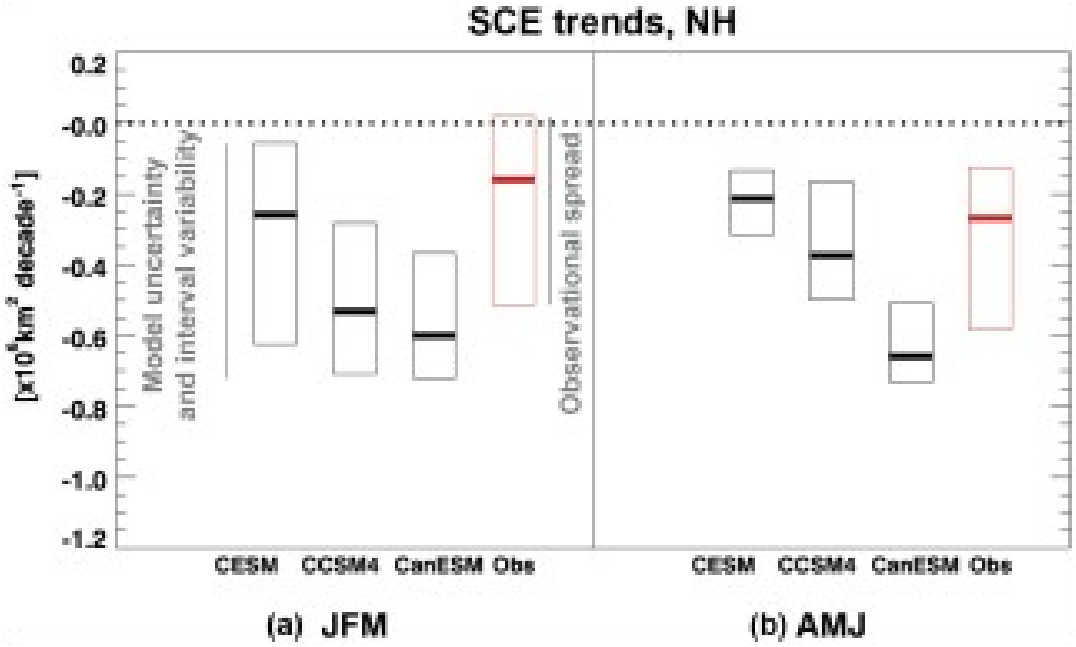


Figure S1.8: Boxplot showing the median and interquartile range of northern hemispheric snow cover extent trends calculated over 1981–2005 period for the CESM-LE, CCSM4-LE, and CanESM2-LE, as well as for the observations, over January–March (a) and April–June (b). *Figure 4 in Kushner et al. (2018).*

MODULE S2

Perturbed Parameter Ensembles

Learning outcomes

1. Explain what perturbed parameter ensembles are, and how they are used to explore parametric uncertainty in climate models.
2. Discuss how model tuning is achieved using model parameters.
3. Explain how cloud and convection parameterizations impact model uncertainty.
4. Describe how statistical emulation is used to quantify parametric uncertainty.

S2.1 Perturbed parameter/physics ensembles

Recall from Section 1.6 that models contain many simplified representations of physical processes — parameterizations. They serve two roles:

- representing processes occurring at scales much smaller than the model grid (convection, cloud and precipitation processes);
- representing the relevant physical effects of Earth system processes such as vegetation and ice.

Many of these parameterizations are **highly uncertain**:

- Their values are often statistical relationships based on limited empirical knowledge.
- They have historically used a *fixed* value, or a narrow range of values, across the whole model.

S2.1.1 Why emulation is necessary

Running the fully coupled model is computationally expensive and time consuming, so exhaustively sampling parameter space directly is not viable.

- Instead we rely on **statistical emulation**: training a statistical model to approximate the behaviour of the complex model.
- An emulator can rapidly explore the sample space created by hundreds to thousands of different parameter combinations.
- The goal is to quantify the proportion of variability attributable to each parameter, identifying which parameters the climate system is most sensitive to.

S2.2 Model tuning and calibration

Tuning

Varying model parameters in such a way that the model output — typically the simulated historical climate — matches observations. The practice is long-established in hydrological modelling.

Models contain parameters that are often uncertain and poorly constrained, and over much of the CMIP era the tuning process was **poorly documented**. That opacity is itself part of

the uncertainty problem: if the tuning targets are not reported, the resulting agreement with observations cannot be independently interpreted.

Figure S2.1 shows that the ensemble mean, global mean temperatures of CMIP3 and CMIP5 models were relatively close to observed estimates from HadCRUT3v and CRUTEM3v. However, the *vast majority of individual models* fail to capture the observed historical temperature trend, with many showing a cold bias.

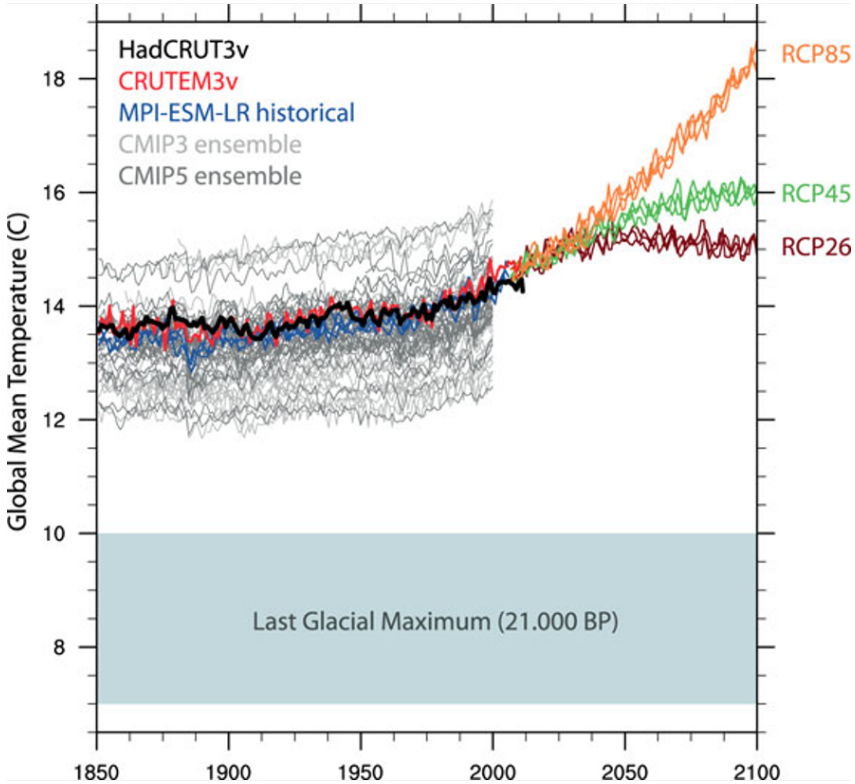


Figure S2.1: Spread in the CMIP3 and CMIP5 ensemble global mean temperature estimates, alongside the RCP2.6, RCP4.5, and RCP8.5 scenarios in the MPI-ESM-LR model. *Figure 1 in Mauritsen et al. (2012).*

Clouds and convection are among the most poorly constrained and uncertain processes in climate models (Figure S2.2). Model tuning requires an understanding of how the radiation balance is altered in response to changes in different parameters. This is often achieved through perturbed parameter ensembles, where the influence of thousands of parameter combinations can be explored (Figure S2.3).

S2.3 Parametric uncertainty

Parametric uncertainty affects the TCR, for example through changes to aerosol forcing and climate sensitivity. **Differently tuned versions of the same ESM can produce very different 20th century warming trends.**

S2.3.1 Cloud droplet parameterizations

Golaz et al. (2013) showed that the default GFDL-CM3 model tended to under-predict the warming over the historical period owing to an overly strong cooling from aerosol effects (Figure S2.4). They tested the influence of two alternate configurations of the GFDL-CM3 model in which the threshold cloud droplet size was altered:

- **CM3w** uses a *lower* threshold cloud droplet size than the default model — a less physically realistic value.

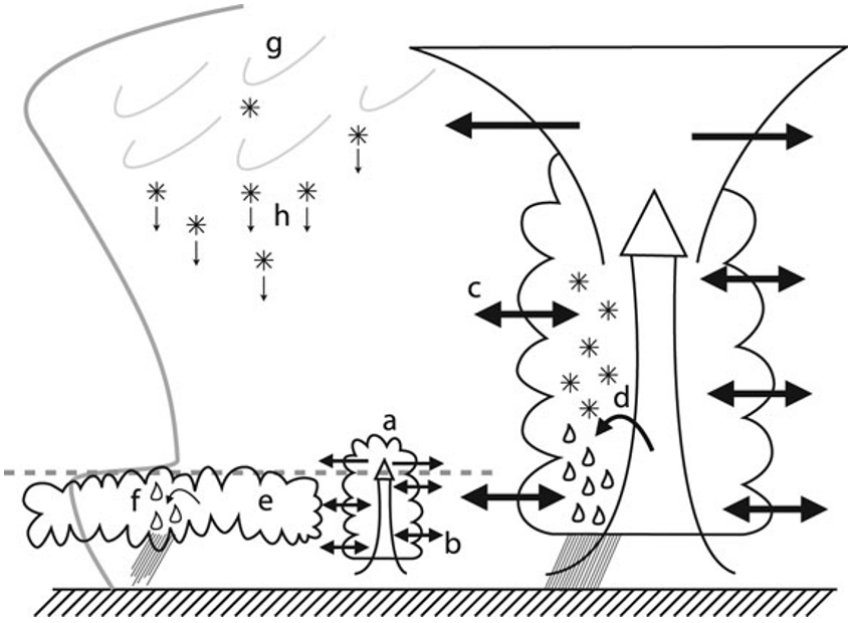


Figure S2.2: Illustration of the major uncertain climate-related cloud processes frequently used to tune the climate of the ECHAM model, including a) convective cloud mass-flux above the level of non-buoyancy, b) shallow convective cloud lateral entrainment rate, c) deep convective cloud lateral entrainment rate, d) convective cloud water conversion rate to rain, e) liquid cloud homogeneity, f) liquid cloud water conversion rate to rain, g) ice cloud homogeneity, and h) ice particle fall velocity. *Figure 2 in Mauritsen et al. (2012).*

- **CM3c** uses a *higher* threshold cloud droplet size — a more desirable value — along with a reduced lower bound on velocity variance for activation, to offset the impact of the higher threshold.

Table S2.1 shows the global mean warming trend between 1981-2000 for the three model configurations. The configuration which reproduces the observed warming trend most closely is CM3w, in which the threshold cloud droplet size was reduced relative to the default model. Noteably, CM3c, which uses the most realistic cloud droplet threshold size, produces an even larger cold bias. This illustrates the sensitivity of the transient climate response (TCR) to parametric uncertainty, and shows how tuning can sometimes take advantage of compensating errors to produce the "correct" global mean temperature trend for the wrong reasons.

Configuration	1981–2000 warming	Parameter realism
CM3 (default)	0.22 °C	Default
CM3w	0.57 °C	Less realistic (lower threshold)
CM3c	−0.01 °C	More realistic (higher threshold)

Table S2.1: Warming relative to 1881–1920 across three GFDL CM3 configurations. The two modified versions span 0.58 °C, and CM3w — the version using the *less* realistic parameter value — is closest to the observed estimate.

S2.3.2 Ice crystal fall velocity

Hourdin et al. (2017) note that GFDL-CM3, MPI-ESM1.1, IPSL-CM5A and IPSL-CM5B share similar parameterizations for ice crystal fall velocity, albeit with different scaling factors (Figure S2.5a):

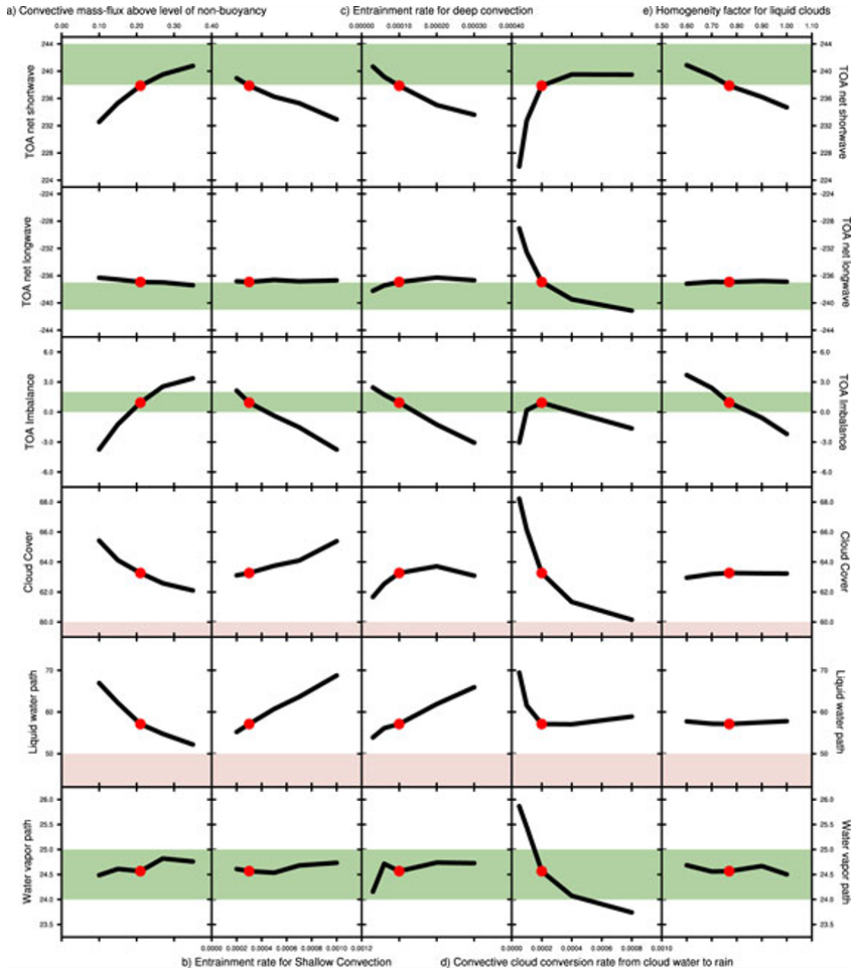


Figure S2.3: The response of globally averaged cloud properties in the ECHAM6 atmospheric module of MPI-ESM-LR to five poorly constrained model parameters. Green bands show the best observational estimates; pink shows ranges considered "unacceptable"; red dots are the values obtained from the default ECHAM6 model using prescribed SSTs and sea ice. *Figure 3 in Mauritsen et al. (2012).*

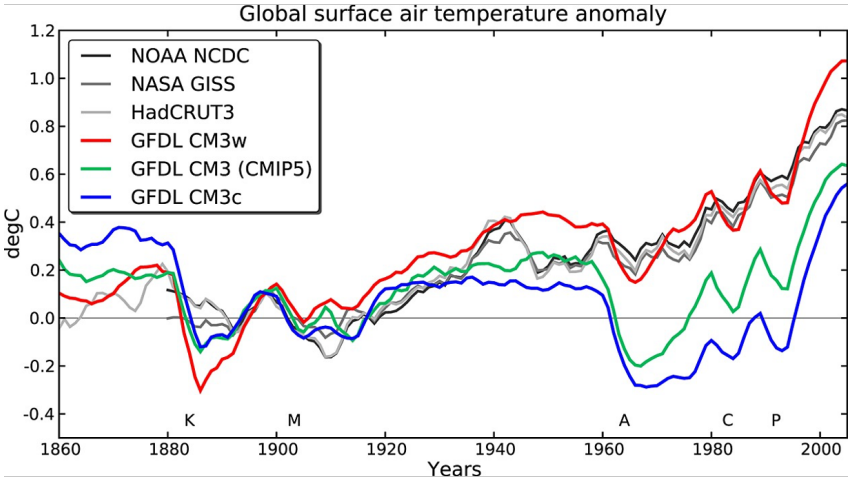


Figure S2.4: Global surface air temperature anomaly (relative to 1881–1920) in three configurations of the GFDL CM3 model. The default version is shown in blue; CM3w and CM3c are two configurations in which the threshold cloud droplet size has been altered. *Figure 4 in Golaz et al. (2013).*

- All four models share *similar* parameterizations of ice crystal fall velocity, yet produce **substantially different** estimates of the TOA energy balance for a given fall velocity.
- It is clear that fall velocity has been **tuned** in each model in order to bring the TOA energy balance within observational estimates.
- Figure S2.5b,c show how tuning fall velocity in the two IPSL versions affects the zonal mean TOA energy balance, with tuned versions shown as thick lines.

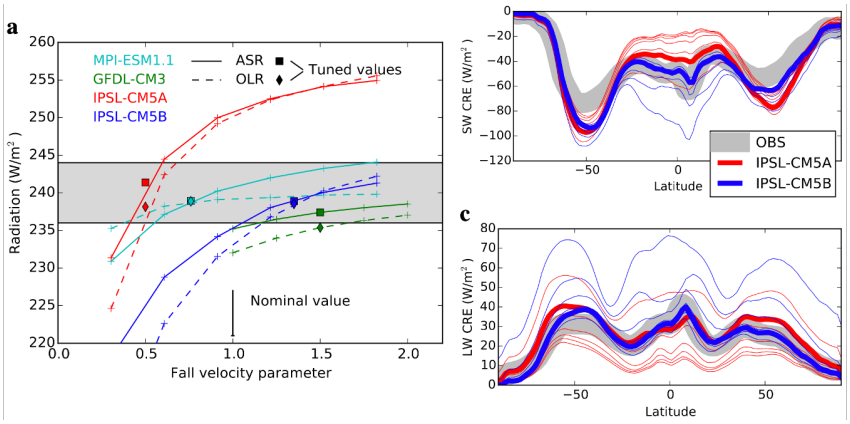


Figure S2.5: a) Global TOA energy balance as a function of ice crystal fall velocity in GFDL-CM3, MPI-ESM1.1, and two versions of IPSL-CM5 (differing in their representations of convection and cloud cover). Absorbed shortwave radiation (ASR) is the solid line, outgoing longwave radiation (OLR) the dashed line. The grey band is the observational estimate of the TOA energy balance; tuned values of ice crystal fall velocity are marked with squares and diamonds. b) SW cloud radiative effects and c) LW cloud radiative effects as a function of latitude in the IPSL-CM5A and IPSL-CM5B models. *Figure 2 in Hourdin et al. (2017).*

S2.3.3 Convection

Cloud base updraft mass flux in a simple column model is parameterized as a function of the CAPE (convective available potential energy, F) consumption rate, using a time scaling factor τ. The time scaling parameter τ is **very poorly constrained**, with estimates ranging between 1800 s and 28 800 s — a factor of 16 (Figure S2.6).

It is shown in section 2 that the large-scale temperature and moisture changes in both the cloud layer and sub-cloud layer are linearly proportional to the cloud base updraft mass flux M_b. Therefore, the CAPE change due to convection can be symbolically written as:

$$\left(\frac{\partial A}{\partial t}\right)_c = -M_b F \qquad (15)$$

where F is the CAPE consumption rate per unit cloud base updraft mass flux.

The closure condition in this study is that CAPE is consumed at an exponential rate by cumulus convection with characteristic time scale τ. This assumption implies that

$$M_b = \frac{A}{\tau F} \qquad \tau \sim 3,600\ s\ (1,800 \rightarrow 28,800\ s)$$

Sensitivity to the choice of the time scale is discussed in the following section in the context of a simple single-column model.

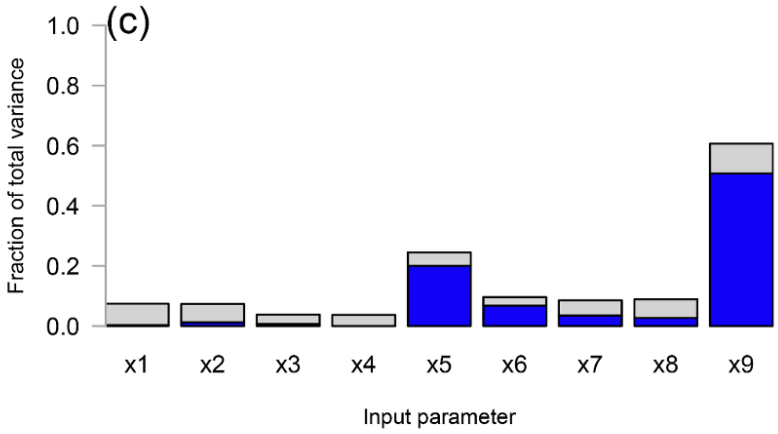
Figure S2.6: The parameterization of the CAPE consumption rate in Zhang and Mcfarlane (1995).

S2.3.4 Statistical emulation of CAM4

Fletcher et al. (2018) explored parameter tuning in the CAM4 atmospheric module of CCSM4, and its effect on the Cess climate sensitivity, using prescribed SSTs with a uniform 2 K warming. Recall from Equation (2.8) that the Cess climate sensitivity, λ_{Cess} is:

$$\Delta T_{\text{Cess}} = \frac{-F_{2\times}}{\lambda_{\text{Cess}}}. \tag{S2.1}$$

They used a statistical emulator of the CAM4 atmospheric module to explore the impact of nine aerosol and cloud/convection related parameters, including the time scaling parameter τ (default CAM4 value 3600 s), shown in the figure as x_9. It was found that variations in the time scaling parameter τ alone accounted for **$\sim$60 % of the variance** in the Cess climate sensitivity (Figure S2.7).



$\tau \sim 3,600\ s\ (1,800 \rightarrow 28,800\ s)$

Figure S2.7: Fraction of total variance explained by λ_{Cess}. The light gray portion of each bar represents the contribution of interaction terms, while the blue portion represents the main effect. The full height of each bar represents the total variance explained by each input parameter, with more influential parameters explaining a higher fraction of variance. *Adapted from Figure 4 in Fletcher et al. (2018).*

MODULE S3

Emergent Constraints

Learning outcomes

1. Understand what emergent constraints are, and how they are used to constrain uncertainty in future climate projections.
2. Discuss how the relationship between the strength of the snow albedo feedback (SAF) over the observed seasonal cycle and the strength of SAF in future projections can be used as an emergent constraint.
3. Explain how cloud albedo feedbacks have been used to constrain estimates of equilibrium climate sensitivity.

S3.1 What are emergent constraints?

An **emergent constraint** exploits a relationship, found across a model ensemble, between something we *can* observe today and something we *cannot* — a future response (Figure S3.1). If the relationship is robust, an observation of the present narrows the range of plausible futures.

The relationship between an observable quantity X and an unknown future response Y is modelled as:

$$Y = f(X) + \varepsilon. \tag{S3.1}$$

where

X observable predictor, measurable over the historical record
Y predictand, the future response we wish to constrain
f relationship diagnosed across the model ensemble, usually assumed linear
ε residual scatter about that relationship

The method is then:

1. Diagnose f across the model ensemble — each model contributes one (X, Y) pair.
2. Find the intersection of the *observed* estimate of the predictor X with the ensemble relationship.
3. Read off the constrained range of Y, which is narrower than the raw ensemble spread.

S3.1.1 Assumptions and conditions

Emergent constraints rely on several key assumptions about the relationships between X and Y:

- Errors must affect model X and model Y in **similar ways**, so that constraining X genuinely constrains Y.
- The model ensemble should ideally include a large number of **independent** samples. This is difficult in practice, given the shared components between models discussed in Module 4.
- The range of X across the ensemble should be **much larger** than the observational uncertainty in X.
- The real-world X should fall **within** the range of model values of X.

- There should be a large range in Y across the ensemble, and that range should exceed the individual model uncertainties.

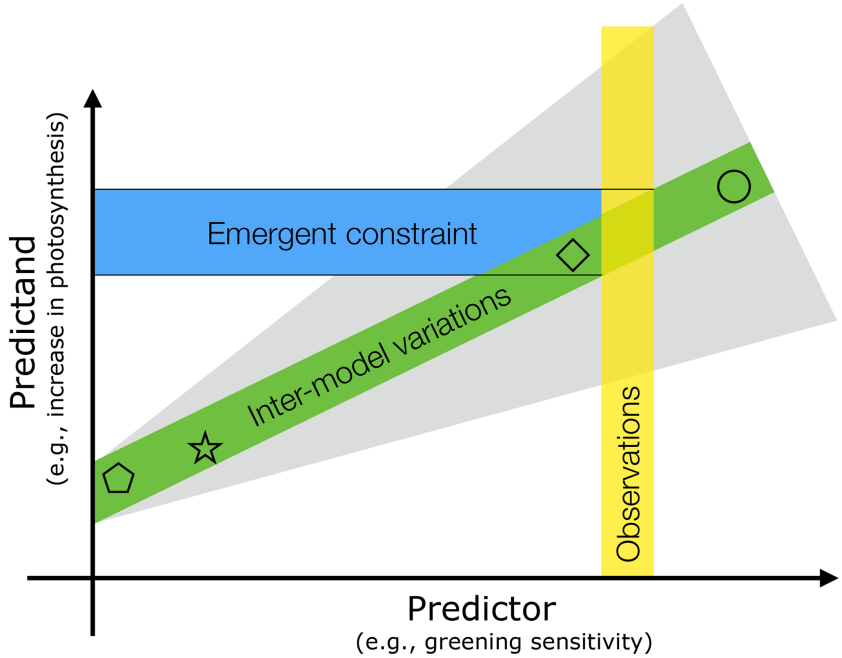


Figure S3.1: Schematic depiction of the emergent constraint (EC) method and factors affecting the uncertainty of the constrained estimate. The predictor (x axis) is change in the annual maximum of green leaf area index (LAImax) due to unit forcing (CO2 increase and associated climatic changes) during a representative historical period. *Figure 1 in Hourdin et al. (2017).*

S3.1.2 Caveats

A number of caveats exist regarding emergent constraints. First, statistically significant regressions **may not be physically based**. Finding a statistically significant correlation across ~30 models is easy to find by chance. Second, missing processes in ESMs could be skewing the relationship, leading to inaccuracies in the true constraint. Third, emergent constraints typically assume a **linear** relationship between X and Y, and it is possible that the relationship between X and Y could be very different in a future climate, particularly if tipping points are crossed. Fourth, the ensemble relies on interrelated models that are not truly independent samples, as many models share inter-related paramterizations and components. Finally, **There is no easy way to verify the relationship**, as Y is an unobservable quantity, relying on model simulations of a future climate.

The central tension

An emergent constraint is a statistical relationship used to make a physical claim about a future we cannot observe, diagnosed from an ensemble we know is not independent. That does not make the technique invalid — it makes physical justification of the relationship, rather than its correlation coefficient, the thing to scrutinise.

S3.2 Snow albedo feedback as an emergent constraint

S3.2.1 Snow Albedo Feedback and State dependence

State dependence refers to the fact that the strength of a feedback may vary under different climate states — which could violate one of the key assumptions above. Wetherald and Manabe (1975) showed that the surface temperature dependence on incoming solar radiation is not constant, and decreases as the snow covered area shrinks (and incoming solar radiation increases). Moreover, Levis et al. (2007) found that in CMIP3 models, higher surface albedo was typically associated with greater snow cover and a stronger SAF (Figure S3.2a), as well as greater warming over higher latitudes (Figure S3.2b), and a larger global mean warming (Figure S3.2c). Since SAF is sensitive to the fraction of snow cover in a reference climate state, the strength of SAF under current climate change may not be the same as its strength during the last glacial period.

S3.2.2 Snow Albedo Feedback

Hall and Qu (2006) first showed that snow albedo feedback (SAF) **over the seasonal cycle** could be used as an emergent constraint on the strength of SAF under future climate change. The relationship between SAF over the seasonal cycle, and SAF under future climate change has remained **remarkably consistent** across CMIP3, CMIP5, and CMIP6 (Figure S3.3). That persistence across three model generations is the strongest available evidence that the relationship is physical rather than coincidental. Using this relationship, Thackeray et al. (2021) suggest the most likely SAF strength under future climate lies between $0.8\,\mathrm{W\,m^{-2}\,K^{-1}}$ and $1.24\,\mathrm{W\,m^{-2}\,K^{-1}}$ (Figure S3.3).

S3.3 Emergent constraints and the role of clouds

There have been sustained attempts to develop emergent constraints on ECS. See, for example Caldwell et al. (2018). Recall that in the classical decomposition of Module 3, ECS was compared against a variety of feedbacks: albedo, lapse rate, water vapour, Planck, and cloud feedbacks. Caldwell et al. (2018) found a strong correlation with ECS for many emergent constraints (Figure S3.4, with **cloud feedback being the dominant contributor** to uncertainty in ECS in both CMIP3 and CMIP5 (Figure S3.4a), and this continues to be true in CMIP6. In CMIP5, cloud feedbacks were decomposed into both shortwave (related to cloud albedo) and longwave components (related to temperature changes), and it was found that the shortwave cloud feedbacks were dominant (Figure S3.4b).

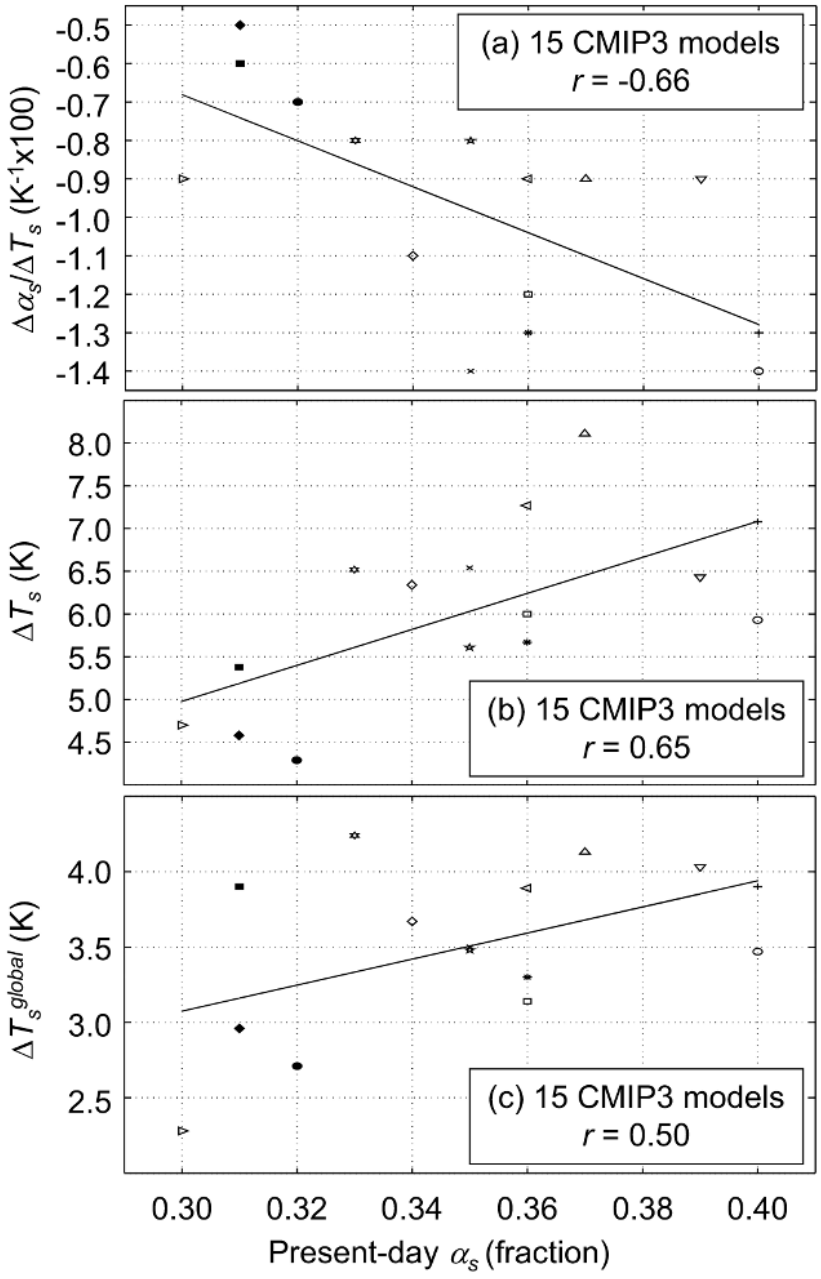


Figure S3.2: Schematic depiction of the emergent constraint (EC) method and factors affecting the uncertainty of the constrained estimate. The predictor (x axis) is change in the annual maximum of green leaf area index (LAImax) due to unit forcing (CO2 increase and associated climatic changes) during a representative historical period. *Figure 1 in Levis et al. (2007).*

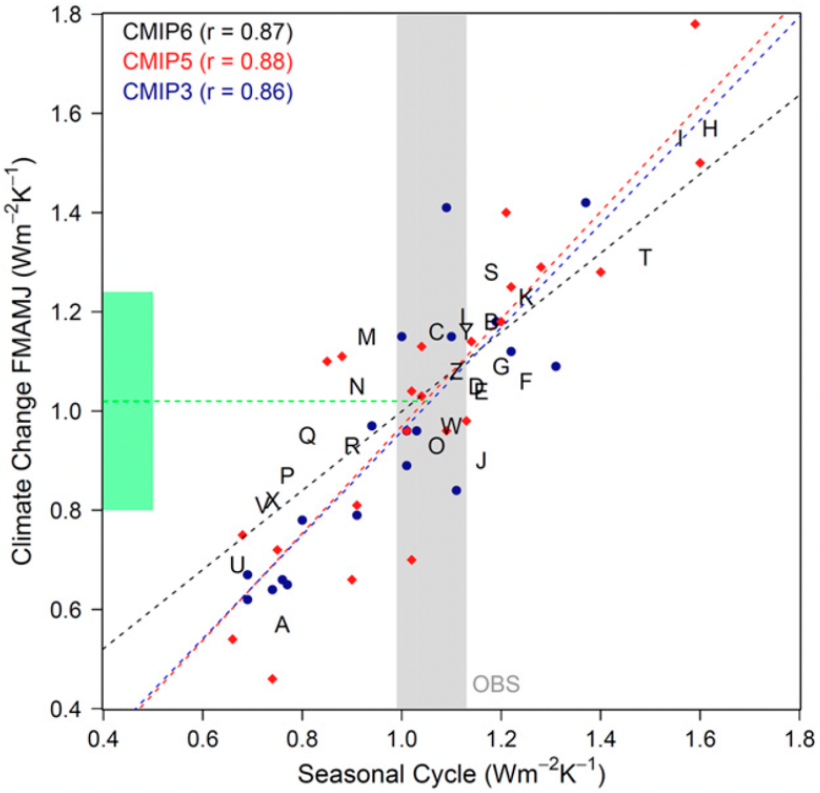


Figure S3.3: Scatterplot of snow albedo feedback in the seasonal cycle (mean of all month-to-month transitions during February–July) context and under climate change (February–June seasonal mean). Each CMIP6 model is shown by a letter (corresponding to Table 1), while red diamonds (23 in total) and blue circles (19 in total) correspond to CMIP5 and CMIP3 models, respectively. *Figure 4 in Thackeray et al. (2021).*

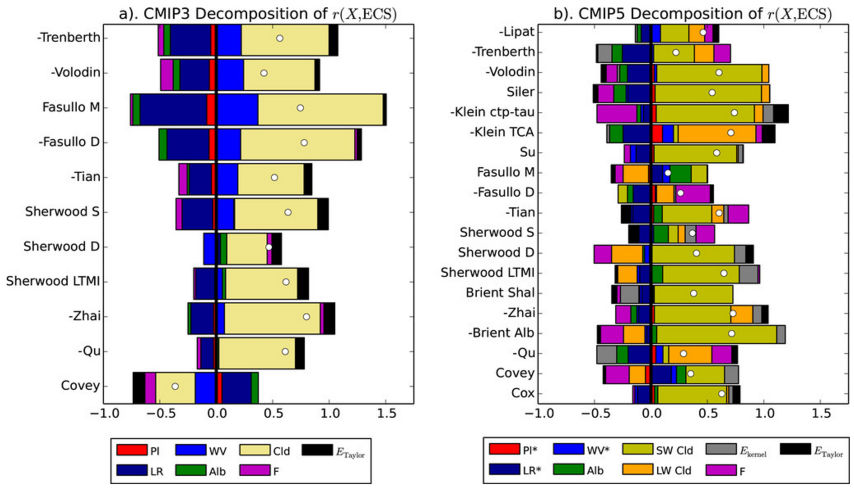


Figure S3.4: Correlation of ECS against each classical feedback term, with the proposed emergent constraints from the literature listed on the y-axis. The cloud feedback term dominates. *Figure 4 in Caldwell et al. (2018).*

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